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Can investors anticipate post-IPO mergers and acquisitions?☆

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ABSTRACT

Given the frequency and value implications of post-IPO merger and acquisition activity, we investigate empirically whether investors can utilize information based on IPO deal structure to predict merger and acquisition activity among newly public firms. Consistent with the hypothesis that some firms conduct IPOs to facilitate future M&A activity, we find that aspects of IPO deal structure predict whether a newly public firm subsequently becomes a bidder or target. These characteristics include underwriter quality, promotional activity, pricing, proceeds, ownership structure, and issuance activity suggestive of market timing. Investors appear to rely on these observable aspects of a firm's going public process to anticipate the implications of M&A activity for security valuation. Specifically, when newly public firms with IPO deal structures predictive of acquisition activity announce an acquisition their stock returns are indistinguishable from zero. In contrast, abnormal returns to acquisition announcements by unlikely or surprise bidders are positive on average. These results suggest that the going public process has important implications for future M&A activity and valuation.

1. Introduction

Facilitation of future merger and acquisition activity is the most prominently mentioned motivation for firms to go public, and newly public firms engage in M&A at a torrid pace shortly after their initial public offerings (Brau and Fawcett, 2006). For example, among nearly 6000 US firms with IPOs between 1980 and 2008, we observe that 38% become merger bidders within three years of the IPO and that 12% become takeover targets in our sample. M&A activity is an important economic fundamental for newly public firms, as on average the dollar volume of post-IPO acquisition activity exceeds the resources devoted to capital investments and R&D expenditures (Celikyurt et al., 2010). Post-IPO merger and acquisition activity also has important implications for post-IPO stock-price performance, especially among newly public bidder firms (Brau et al., 2012; Ritter, 2015). Given the frequency and its important value implication of post-IPO M&A activity, it is critical for investors to recognize IPO firms' tendency to acquire or to be acquired. If investors have the ability to distinguish potential acquirers and targets from firms which do not get involved with the M&A market, then they may tilt their exposure toward or away from firms based on their likely future M&A activity.

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Two examples illustrate that investors can discern likely future M&A activity for some IPO firms. First Solar, the second largest maker of solar panels worldwide, explicitly disclosed that a primary use of its 2006 IPO proceeds was to engage in acquisitions to achieve vertical integration. Not surprisingly, First Solar acquired Turner Renewable Energy >1 year after going public. In contrast, eBay approached online payments servicer PayPal as an acquisition target well before the latter's 2002 IPO, and eBay acquired PayPal shortly after its IPO. Investors probably had little trouble in discerning the nature of likely future M&A activity for First Solar and PayPal after they went public, but what about other newly public firms – many with more ambiguous public information about possible post-IPO M&A activity? Do readily observable aspects of the IPO process suggest which firms may soon become bidders and which firms targets? Furthermore, do security price reactions to subsequent M&A announcements for newly public firms suggest that investors utilize such information in pricing equity securities? These are the question we address in this study.

We propose *the M&A anticipation hypothesis* that investors anticipate post-IPO mergers and acquisitions. This hypothesis is tested in three steps. First, we investigate whether investors can utilize the observable aspects of IPO deal structure to make inferences about the likelihood and nature of future M&A activity. Certain IPO deal structures and IPO outcomes may be conducive to and therefore predictive of whether a firm will eventually bid to acquire another firm, become a merger target, or not become involved in any M&A activity. This perspective is consistent with Brau and Fawcett (2006) who identify facilitation of acquisitions as the most prominent motivation for a firm to go public. Our approach and findings also complement and add to prior empirical studies that link the IPO to subsequent merger activity (Celikyurt et al., 2010; Hovakimian and Hutton, 2010). In contrast to these studies, we consider three possible outcomes for the post-IPO firm – acquire, be acquired, or engage in no M&A activity. We also expand the set of IPO-specific variables beyond pricing and proceeds as elements of IPO structure that might reveal or influence whether a given firm intends to acquire or to be acquired. Our methods and findings complement those of Field and Karpoff (2002) and Boulton et al. (2010) who suggest that soon-to-be public firms choose antitakeover provisions in corporate charters and influence IPO pricing to facilitate or impede becoming a target.[1]

Second, we examine whether institutional investors take stakes in newly public firms based on information observable at the time of the IPO. Field and Lowry (2009) suggest that institutional investors appear skilled at using public information to identify IPOs likely not to underperform subsequent to the IPO. We investigate whether institutional investors seemingly use public information to predict and take stakes in newly public firms likely to engage in M&A after going public.

Third, we investigate whether investors incorporate the likelihood of post-IPO mergers and acquisitions into valuation when the deals materialize. If investors utilize these observable aspects of IPO deal structure to predict future M&A activity then eventual M&A activity by a newly public firm should be partially anticipated by investors, much like investors foresee acquisitions by firms that are frequent acquirers, announce acquisition programs, or are in an industry experiencing a merger wave among rival firms. The empirical implication of investor anticipation is that newly public firms with IPO deal structures predictive of M&A activity should experience attenuated stock returns when mergers are eventually announced (Schipper and Thompson, 1983; Malatesta and Thompson, 1985; Loderer and Martin, 1990; Cai et al., 2011).

Our main empirical results support *the M&A anticipation hypothesis*. Aspects of the going public process predict the likelihood, timing, and valuation consequences of post-IPO M&A activity. In particular, we find that deal structure variables such as pricing, structure of proceeds, post-IPO ownership structure, and underwriting and promotional efforts help to predict whether a newly public firm becomes an acquisition bidder versus an acquisition target within 3 years of its IPO. The predictive ability of these IPO deal structure variables remains significant when we control for other variables indicative of M&A activity, such as firm-specific financial structure and performance, industry classification, year fixed effects, general equity market conditions, and periods of active M&A markets. We also show evidence complementary to *the M&A anticipation hypothesis* when we examine market-timing aspects of the going public process. Specifically, extant evidence suggests that firms may respond to hot equity markets or merger waves in timing equity issuance (Lowry, 2003; Helwege et al., 2007; DeAngelo et al., 2010). Our results suggest that firms with deal structures conducive to becoming merger bidders appear more likely to time their IPOs opportunistically in response to industry-specific M&A waves and to engage in takeovers relatively quickly after their IPOs. In contrast, firms with deal structures typical of takeover targets do not appear to time their IPOs in response to hot markets or merger waves, and acquisitions of such firms occur with a greater time lag relative to the IPO.

Second, we investigate how investments in likely M&A participants vary across institutions skilled at gathering and utilizing firm-specific information. First, we find that publicly available information seems to explain institutional ownership in newly public M&A participants, and institutional ownership attributable to non-public information or public information that our analysis omits has little power to predict post-IPO M&A activity. To shed more light on institutional investment in IPO firms, we match

[1] Our goal is not to test or refute any particular theory about why firms go public but rather to show that public information at the time of an IPO is predictive of future M&A activity. Rather than unravelling the direction of causality, we put ourselves in the shoes of investors and develop a so-called "early warning model" which predicts future M&A activities of IPO firms based on observable associations and information revealed at the time of IPO. We find that inherent IPO characteristics help predict newly public firms' future M&A activities approximately 70% of the time. Because the focus of interest in our paper differs substantially from prior work in this respect, we take into consideration the likelihood of becoming a bidder, target, or neither, whereas the extant research only distinguishes bidders (e.g., Celikyurt et al., 2010; Hovakimian and Hutton, 2010) or targets (e.g. Zingales, 1995; Boulton et al., 2010) firms, but not both. The closest paper to our study in terms of orientation, Bharath and Dittmar (2010), investigates the predictability of public firms going private and acknowledge that inherent firm characteristics observed at the time of IPO significantly reveal discernible information about firms which, on average, remain in the public market for over 13 years, but eventually go private. However, Bharath and Dittmar (2010) do not examine M&A activities. Our paper also contributes to the partial anticipation hypothesis in other contexts. For example, future bidding activity can be anticipated and priced into stock returns early for program acquirers and same-industry bidders (Schipper and Thompson, 1983; Malatesta and Thompson, 1985; Cai et al., 2011).

eventual bidder firms to non-bidder firms based on a wide array of firm characteristics *excluding* IPO deal characteristics and find that transient institutional investors hold larger ownership stakes – and larger stakes than so-called dedicated institutional investors – in eventual bidders, compared to non-bidders. Overall, these results indicate that some institutional investors exploit publicly available information available at the time of a firm's IPO to anticipate post-IPO mergers.

Finally, we find that the predictability of post-IPO merger activity has implications for security pricing upon and after merger announcements. Specifically, we find that excess stock returns upon merger announcement are indistinguishable from zero among newly public firms with IPO deal structures predictive of bidding activity, in contrast to positive returns to unlikely or surprise bidders. These results suggest that value-enhancing bidding activity is already capitalized into market equity values among likely bidders, consistent with the investor anticipation hypothesis. These results are inconsistent with the notion that unlikely or ill-prepared newly public bidders make overvalued bids or otherwise fail to create value through M&A activity (Banerjee and Owers, 1992; Brau et al., 2012). However, among unlikely bidders we find evidence of market overreaction to merger announcements, suggesting that investors imperfectly process the somewhat surprise news of a merger bid by an unlikely bidder.

The remainder of this study is organized as follows. In Section 2, we identify our sample and then develop our empirical hypotheses. In Section 3, we estimate how IPO characteristics predict M&A activity among newly public firms. Section 4 examines investment behavior of institutional investors at likely and unlikely post-IPO bidders. In Section 5, we investigate how market reaction to announcements of M&A activity is conditioned by expectations based on IPO features. Section 6 concludes.

2. Data and methods

In this section we describe our sample of newly public firms and how we observe M&A activity subsequent to the IPO. We then identify and articulate the aspects of IPO deal structure that are conducive to and therefore predict M&A activity involving newly public firms.

2.1. IPO firm sample and post-IPO M&A activity

We examine a comprehensive sample of U.S. IPOs from 1980 to 2008. Our data source is the Thomson Financial Securities Data U.S. Common Stock Initial Public Offerings database (new issues database) with additions and corrections based on Jay Ritter's identification, CRSP, and Compustat. The sample is composed of 5927 IPOs meeting criteria that are common in the empirical IPO literature (e.g. Loughran and Ritter, 2004; Liu and Ritter, 2011).[2] Data for analyst coverage, dual class, and firm founding date are provided by Jay Ritter.[3] Table 1 displays by calendar year the number of IPOs, total issue proceeds (in 2003 inflation-adjusted dollars), mean and median proceeds, and mean and median initial returns observed on the first day of public trading. The sample period spans several so-called “hot” IPO markets characterized by relatively high issue volume and initial returns as well as “cold” IPO markets characterized by low volume and initial returns.

We rely on the SDC M&A database to identify acquisition activity involving sample firms for 3 years following their IPOs. To be considered as a merger the bidder's prior holdings should be <50% of the target's shares, and the resulting share ownership of the acquirer should be >90% (Hovakimian and Hutton, 2010). Acquisitions do not include buybacks, recapitalizations, liquidations, restructurings, leveraged buyouts, reverse takeovers, privatizations, bankruptcy acquisitions, going private transactions, spinoffs, or self-tenders (Celikyurt et al., 2010). We require the deal value to be non-missing in SDC.

Table 1 also shows the frequency at which firms with IPOs from a calendar year are involved in M&A as a bidder or target in the subsequent 3 years. On average, 1 year after the IPO, 21.9% of sample firms become bidders and 2.4% targets. The cumulative incidence of bidder and target activity increases to 31.5% and 6.9%, respectively, for the 2 years after the IPO, and to 37.7% and 11.7% for 3 years after the IPO. The frequencies of M&A activity in our sample are comparable to those reported by Hovakimian and Hutton (2010), Celikyurt et al. (2010), Boulton et al. (2010), and Hsieh et al. (2011). Two patterns are apparent. Foremost, for newly public firms engaging in M&As as a bidder is much more prevalent than becoming a merger target. In addition, across the sample period the frequency of M&A activity varies with the presence of hot versus cold IPO markets as measured by offer frequency and initial returns, consistent with prior literature establishing a link between equity market conditions and so-called waves of security issuance and M&A activity.

The cumulative M&A frequencies we report in Table 1 do not double-count any serial bidding activity. However, some sample firms initially engage in M&A as bidders and then later become merger targets. Fig. 1 displays the cumulative frequency of post-IPO outcomes including delisting for non-merger reasons, becoming a bidder firm, becoming a bidder firm and a later target firm, becoming a target firm without first being a bidder, and the firm surviving for 3 years after the IPO without any observable M&A activity. The cumulative frequency at which a sample firm first bids as an acquirer and later itself becomes a target is 0.56% 1 year after the IPO, 2.21% 2 years after the IPO, and 4.54% 3 years after the IPO.[4]

[2] Specifically, we exclude (a) observations not identified on CRSP, (b) observations in which the offering is not underwritten or is not available, (c) limited partnerships, (d) REITs, (e) closed-end funds, (f) spinoffs, (g) previous leverage buyouts, (h) units, (i) ADRs, (j) shares of beneficial interest, (k) foreign corporations, (l) banks and savings and loans (S&Ls), (m) best efforts offers, (n) SPACs, (o) IPOs with an offer price of less than $5.00 per share, (p) utility firms, and (q) IPOs which are not original. We also report the frequency of an extended sample of 6044 IPOs from 1980 through 2011 in Table 1.

[3] Following Liu and Ritter (2011), the analyst dummy is set to zero for IPOs prior to 1993 when analyst information became available, with the year fixed effects for these years incorporating the effects of the missing information.

[4] Our reported results are not sensitive to inclusion or exclusion of the IPO firms that are initially acquirers and later become targets.

Table 1
IPO sample and subsequent merger activity.

Year	N	IPO Proceeds (millions)			Initial return		IPO + 1 year		IPO + 2 years		IPO + 3 years	
		Total	Mean	Median	Mean	Median	Bidder	Target	Bidder	Target	Bidder	Target
1980	62	$1743.1	$28.1	$17.5	15.8%	10.1%	3.2%	0.0%	6.5%	4.8%	9.7%	6.5%
1981	163	$3915.6	$24.0	$16.7	7.9%	2.7%	6.1%	0.6%	11.7%	2.5%	16.0%	5.5%
1982	61	$1593.9	$26.1	$13.0	12.4%	4.6%	11.5%	0.0%	16.4%	6.6%	19.7%	9.8%
1983	352	$11,941.0	$33.9	$20.3	12.3%	4.1%	9.1%	1.4%	13.1%	4.3%	15.9%	6.0%
1984	136	$2314.1	$17.0	$12.4	4.8%	1.4%	4.4%	2.9%	5.9%	4.4%	7.4%	5.9%
1985	113	$3090.2	$27.3	$18.5	7.0%	3.1%	2.7%	2.7%	6.2%	4.4%	10.6%	8.8%
1986	311	$16,158.0	$52.0	$19.1	8.0%	2.5%	6.8%	1.6%	11.3%	3.9%	16.7%	8.0%
1987	210	$10,527.0	$50.1	$21.2	7.4%	2.1%	5.2%	1.4%	13.8%	4.3%	19.5%	6.2%
1988	84	$4414.9	$52.6	$23.6	6.3%	2.6%	6.0%	1.2%	15.5%	2.4%	21.4%	6.0%
1989	91	$5783.0	$63.5	$25.1	9.3%	4.7%	11.0%	0.0%	26.4%	2.2%	36.3%	5.5%
1990	81	$3796.8	$46.9	$29.0	11.7%	5.6%	22.2%	0.0%	40.7%	1.2%	50.6%	2.5%
1991	200	$10,809.0	$54.0	$37.0	12.7%	8.7%	19.5%	1.5%	31.0%	3.0%	43.5%	5.5%
1992	276	$17,295.0	$62.7	$30.5	10.5%	4.2%	17.8%	0.7%	30.8%	3.6%	39.9%	8.3%
1993	403	$26,274.0	$65.2	$33.5	13.8%	7.1%	22.3%	1.0%	33.3%	5.0%	40.4%	8.9%
1994	330	$16,826.0	$51.0	$29.0	10.1%	4.7%	21.5%	1.8%	37.6%	7.0%	45.5%	12.7%
1995	412	$27,774.0	$67.4	$37.9	22.4%	14.6%	26.9%	3.2%	38.1%	9.5%	45.6%	17.0%
1996	601	$38,912.0	$64.7	$38.1	17.0%	10.0%	29.3%	2.8%	41.4%	9.2%	49.3%	18.3%
1997	388	$25,289.0	$65.2	$36.6	14.4%	10.0%	35.3%	2.3%	45.6%	8.2%	50.0%	14.2%
1998	234	$19,231.0	$82.2	$43.9	20.9%	10.6%	38.0%	3.8%	47.0%	10.7%	54.7%	13.7%
1999	413	$53,241.0	$128.9	$67.4	65.7%	44.3%	46.0%	7.3%	54.2%	16.0%	57.1%	21.5%
2000	309	$42,616.0	$137.9	$80.1	52.5%	27.2%	27.2%	4.2%	37.5%	9.4%	42.7%	14.2%
2001	57	$12,517.0	$219.6	$88.8	15.9%	15.0%	24.6%	1.8%	36.8%	5.3%	45.6%	12.3%
2002	56	$8164.0	$145.8	$92.1	10.1%	8.5%	17.9%	5.4%	33.9%	8.9%	42.9%	17.9%
2003	56	$7541.5	$134.7	$97.8	12.7%	10.1%	25.0%	7.1%	35.7%	8.9%	41.1%	10.7%
2004	152	$19,882.0	$130.8	$75.7	12.3%	7.4%	14.5%	0.7%	26.3%	7.2%	34.9%	13.8%
2005	129	$17,141.0	$132.9	$83.9	11.1%	6.5%	31.0%	2.3%	40.3%	6.2%	43.4%	11.6%
2006	123	$19,616.0	$159.5	$84.4	13.0%	9.1%	16.3%	3.3%	27.6%	8.1%	32.5%	12.2%
2007	110	$14,292.0	$129.9	$81.7	14.2%	7.0%	20.0%	0.9%	24.5%	5.5%	33.6%	10.9%
2008	14	$3482.4	$248.7	$111.2	6.0%	−3.3%	14.3%	0.0%	28.6%	0.0%	50.0%	14.3%
2009	20	$4492.6	$224.6	$117.4	13.5%	6.2%	25.0%	0.0%	25.0%	0.0%		
2010	53	$18,687.0	$352.6	$76.0	9.7%	3.6%	17.0%	3.8%				
2011	44	$5682.6	$129.1	$87.0	13.5%	11.2%						
Total	6044	$475,043.0	$78.6	$38.1	18.7%	7.5%	21.9%	2.4%	31.5%	6.9%	37.7%	11.7%

This table provides the time series of IPO and subsequent merger activity. Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008, and we report the frequency of an extended sample of 6044 IPOs from 1980 through 2011 in this table. We track IPO firms' acquisition activity for up to three years post-IPO. In addition to the number of IPOs, the table depicts the dollar amount of total IPO proceeds from all IPOs in a given year, mean and median values of IPO proceeds in dollars as well as initial first day returns in percentage. Proceeds and assets are expressed in 2003 dollars. All IPO variables are defined in Appendix A.

2.2. Factors that predict post-IPO M&A activity

We hypothesize that readily observable IPO deal structures and outcomes are conducive to and therefore predictive of whether a firm will eventually bid to acquire another firm, become a merger target, or not become involved in any M&A activity. Table 2 shows IPO-specific variables of interest split into the following categories: (i) pricing and proceeds, (ii) ownership structure, (iii)

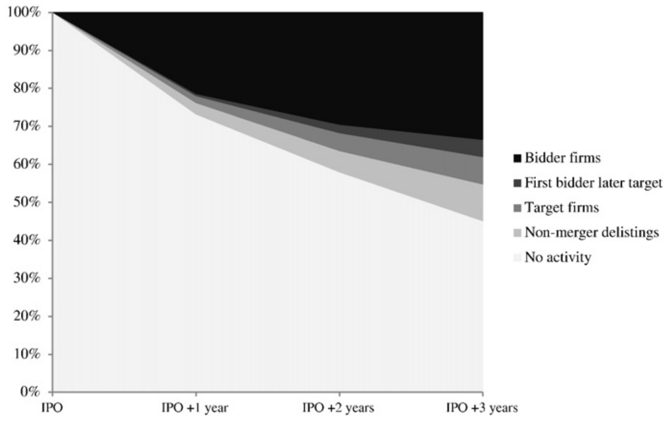

Fig. 1. Cumulative post-IPO merger activity. This figure displays the cumulative frequency of post-IPO outcomes including delisting for non-merger reasons, becoming a bidder firm, becoming a bidder firm and a later target firm, becoming a target firm without first being a bidder, and the firm surviving for 3 years after the IPO without any observable M&A activity.

Table 2
Testable hypotheses for the determinants of becoming a post-IPO bidder versus target.

IPO deal structure	Hypothesis	Variable	Hypothesized effect on bidder likelihood	Hypothesized effect on target likelihood
Pricing & proceeds	Secondary market liquidity, capital infusion, uncertainty resolution, and feedback effect.	*Underpricing*	Positive	N/A
		Log(Primary Proceeds)	Positive	N/A
		Log(Secondary Proceeds)	N/A	N/A
		Log(Overallotment Proceeds)	Positive	N/A
		Price Revision	Positive	N/A
		Disclosure Dummy	Positive	N/A
		Post-IPO 20-day Return	Positive	N/A
Ownership structure	Institutional monitoring, and control retention.	*Overhang*	N/A	Positive/Negative
		Dedicated Institutional	Negative	Positive
		Ownership Ratio	Positive	Negative
		Transient Institutional	Negative	Positive
		Ownership Ratio	Negative	Positive
		IO Breadth	Negative	Negative
		Institutional Ownership HHI		
		Dual Class Dummy		
Promotion & underwriting	Investor belief heterogeneity, and underwriter reputation.	*All-star Analyst Dummy*	N/A	Positive
		Bid-ask Spread Growth	N/A	Positive
		Bookrunner Rank	N/A	Positive
Market condition & growth strategy	Market timing, and growth strategy	*Pre-IPO Sentiment*	Positive	N/A
		Industry Acquisition Intensity	Positive	N/A
		Organic Growth/Assets	Negative	Negative
		Pre-IPO Acquisition Dummy	Positive	N/A
Control		*VC Dummy*		
		Age		
		Financial Ratios (Book Leverage, ROA, Market-to-Book)		
		Industry Dummies		
		Year Dummies		

This table lists four categories of IPO deal structure variables, how they are related to the hypotheses to be tested, and the expected coefficient signs in the following estimations of the likelihood of becoming a bidder or target. All IPO variables are defined in Appendix A.

promotion and underwriting, (iv) indicators of equity market and firm-growth strategies, and (v) additional control variables such as firm age, venture capital-backed status, leverage, profitability, and market valuation. We also include industry- and year-fixed effects to account for industry-specific and time-period specific waves of M&A activity displayed in our sample. Table 2 also summarizes the hypothesized effects of the IPO-specific variables on the likelihood that the firm is going to become a merger bidder or a merger target. These hypothesized effects are discussed in greater detail below.

Appendix A provides detailed variable definitions and data sources, and Table 3 provides sample statistics for subsamples by post-IPO M&A activity, i.e., post-IPO non-merger firms, post-IPO targets, and post-IPO acquirers. Due to data availability for some of our variables of interest and our focus on post-IPO M&A activity within the first one-to-three years following the IPO, our sample sizes that range from around 6000 firms to slightly fewer than 3000 firms depending on the specification.

2.3. Proceeds and pricing

Cash raised through an IPO can be a critical asset for a strategy of growth by acquisition, and IPO proceeds have an enduring effect on the volume of post-IPO acquisition activity (Celikyurt et al., 2010). Consequently, firms with larger cash infusion are more likely to grow via acquisitions. In contrast, firms that view the IPO as a step toward being acquired as targets are expected to be less concerned about proceeds and more concerned about diversifying private owners' shares throughout an investor base and increasing the chance of selling out (Bodnaruk et al., 2008). We measure each IPO's proceeds in inflation-adjusted real dollars, and distinguish the primary proceeds from new shares and secondary proceeds from shares held by pre-IPO shareholders as well as the dollar amount of green-shoe option exercised. The measures are transformed by the natural log in our empirical specifications.

On average, IPOs appear to be underpriced as the post-offer market share price tends to exceed the offer price, and thus initial return is unusually positive. How an IPO is priced has several implications for whether a newly public firm will participate in the M&A market. Likely bidders are particularly interested in raising capital as well as successfully launching the public shares as a potential currency in the secondary market to be used for future acquisitions. We expect that firms which promote underpricing more than other newly public firms are more likely to be post-IPO bidders. By doing so, these firms attract a broader array of investors, issue extra shares via the overallotment option, create broad and dispersed initial ownership, and thereby enjoy liquid acquisition currency. The resulting diffuse ownership may also reflect a firms' defensive motive to remain independent (Brennan and Franks, 1997; Boulton et al., 2010). At the same time, underpricing as a way to promote oversubscription in turn increases secondary-market liquidity (Booth and Chua, 1996). We argue that

Table 3
Descriptive statistics.

Variable	N	Mean	Median	Dev	Min	Max	Non-merger (N = 3243) Mean	Target (N = 427) Mean	Acquirer (N = 1988) Mean	Acq-Tar (N = 273) Mean
Underpricing (%)	5927	18.81	7.50	34.02	−12.65	203.75	14.84	19.81	23.65	29.38
Primary Proceeds ($mil)	5927	56.19	30.95	83.94	0.00	856.20	49.50	57.18	64.96	70.54
Secondary Proceeds ($mil)	5927	11.11	0.00	73.75	0.00	2664.44	9.90	7.63	14.10	9.22
Overallotment ($mil)	5927	8.80	4.10	24.04	0.00	906.19	7.42	7.59	11.02	10.92
Price Revision (%)	4442	−0.75	0.00	18.35	−43.33	63.64	−2.42	−0.44	1.20	4.44
Disclosure Dummy	5927	0.12	0.00	0.33	0.00	1.00	0.09	0.17	0.16	0.16
Cumulative Excess Return	5926	0.02	0.00	0.20	−0.41	0.90	0.01	0.02	0.04	0.07
Overhang (%)	5927	1.81	1.39	2.59	0.00	21.40	1.94	1.76	1.60	1.81
IO Ratio	5927	0.15	0.12	0.14	0.00	0.82	0.13	0.16	0.17	0.19
Dedicated IO Ratio	5865	0.02	0.00	0.03	0.00	0.20	0.02	0.02	0.02	0.02
Transient IO Ratio	5865	0.06	0.04	0.07	0.00	0.32	0.05	0.07	0.07	0.09
IO Breadth	5909	17.74	14.00	18.01	0.00	237.00	14.92	18.90	21.20	24.12
Institutional Ownership HHI	5909	0.20	0.13	0.23	0.00	1.00	0.22	0.20	0.17	0.16
Dual-class Shares Dummy	5927	0.06	0.00	0.25	0.00	1.00	0.06	0.04	0.08	0.06
All-star Analyst Dummy	5927	0.11	0.00	0.32	0.00	1.00	0.09	0.12	0.13	0.20
Bid-ask Spread Growth (%)	4567	17.44	12.44	35.60	−99.84	274.69	17.24	18.47	17.43	17.85
Bookrunner Rank	5927	7.12	8.00	2.18	0.00	9.00	6.83	7.51	7.41	7.80
Pre-IPO Sentiment	5927	0.24	0.05	0.53	−0.83	2.26	0.32	0.22	0.12	0.13
Industry Acquisition Intensity	5916	0.09	0.07	0.07	0.00	0.60	0.07	0.10	0.11	0.13
Organic Growth	5218	0.14	0.11	0.13	0.00	0.67	0.16	0.15	0.12	0.11
Pre-IPO Acquisition Dummy	5927	0.03	0.00	0.18	0.00	1.00	0.02	0.03	0.06	0.06
VC Dummy	5927	0.44	0.00	0.50	0.00	1.00	0.40	0.54	0.47	0.52
Book Leverage	5281	0.16	0.05	0.21	0.00	0.85	0.16	0.14	0.16	0.18
ROA	5288	−0.04	0.03	0.23	−1.11	0.27	−0.05	−0.06	−0.03	−0.04
Market-to-Book	5295	3.28	2.37	3.08	0.78	20.50	2.96	3.40	3.67	3.81
Age	5890	13.58	7.00	18.63	0.00	165.00	13.69	12.32	13.72	13.29
Assets	5306	265.92	72.70	729.18	2.47	8304.80	255.84	255.51	279.40	297.49

Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008, and we track IPO firms' acquisition activity for up to three years post-IPO. The table shows the summary statistics for the full sample. The mean values of the variables in different subsamples, i.e., non-merger, target, acquirer, and acquirer-target subsamples, are also provided. Proceeds and assets are expressed in 2003 dollars. All IPO variables are defined in Appendix A.

post-IPO stock liquidity is more important for post-IPO bidders than issuers which are subsequently sold as it will facilitate completion of an acquisition deal by increasing the chance that a potential target is willing to accept the acquirer's stock as merger currency that can be easily convertible to cash.

We further conjecture that reduction in valuation uncertainty via information production is predictive of post-IPO acquisitions. Hsieh et al. (2011) suggest that a private bidder does not know its true valuation, which affects its gain from an acquisition. An IPO reduces valuation uncertainty during the book building process and allows a potential bidder to pursue a more efficient acquisition strategy. We use the price revision as our proxy of information production which is measured as the percentage difference between offer price and the mid-point of the initial offer price range identified in the initial prospectus. Likely bidders may also get feedback on their acquisitions prospective from the IPO market in an attempt to adjust, re-scale, or perhaps time or abandon their acquisition plans. We employ two other proxies, namely disclosure of use of IPO proceeds and post-IPO stock return over the first 20 trading days, reflecting the channels of managers-investors interaction and information production (Jegadeesh et al., 1993). Overall, we hypothesize that the emphasis on capital infusion, secondary market liquidity, and uncertainty resolution are predictive of subsequent acquisitions.

2.4. Ownership structure

Netter et al. (2009) provide a comprehensive overview of the role of internal and external control mechanisms and ownership structure in corporate control. Consistent with this literature, Boulton et al. (2010) and Brennan and Franks (1997) suggest that managers and insiders at firms averse to being taken over will prefer more diffuse ownership structure as that would impede easy accumulation of a toehold by a prospective acquirer. They suggest that underpricing is used as a means of establishing a more diffuse shareholder base tilted away from institutional owners likely to put their shares in play during a takeover attempt. Firms with a more diffused ownership and weak monitoring by some classes of institutional investors are more likely to implement their acquisition plans.

We rely on Thomson Financial's 13F database to identify a firm's institutional investors within one quarter of the IPO date, constructing institutional ownership percentage and breadth following Bushee (2001). We further calculate a Herfindahl index of overall institutional ownership concentration. We also include a dummy for dual class share structure since we anticipate that firms with dual class shares are less likely to become either targets or acquirers as this structure is frequently employed to prevent dilution of control and maintain insider control versus outside shareholders.

We also investigate the relation between the shares retained by pre-IPO shareholders and post-IPO merger activity by including share overhang in the prediction model. Share overhang can possibly reflect two different motivations that affect target likelihood. On one hand, overhang may reflect that some key pre-IPO shareholders such as founders want to retain ownership and control of the firm, and therefore overhang will be negatively related to the probability of being sold. On the other hand, pre-IPO shareholders such as venture capitalists and private equity may be somewhat involuntarily subject to lockup agreements and therefore may be motivated to exit by becoming a target (Field and Hanka, 2001).

2.5. Underwriting and promotional activities

Several aspects of the underwriting and IPO promotional process may map onto future M&A activity. We therefore include variables such as underwriter's rank as per Carter and Manaster and updated by Jay Ritter, whether underwriters provide analyst coverage in the year after the IPO, and a measure of bid-ask spread expansion, defined as the percentage increase from the average quoted bid-ask spread of the first 20 trading days to the average quoted bid-ask spread of the second 20 trading days.[5] We argue that the IPO promotional process augments the chance of post-IPO firm sale. Chemmanur and Yan (2009) find that promotional activity such as advertising in the IPO year has a positive effect on the investor belief heterogeneity, which, as argued by Chatterjee et al. (2012), yields a higher merger premium. Marketing and promotional activities may augment the chance of selling the firm and accelerate the sale of the company.[6] Finally, Alavi et al. (2008), Brau and Fawcett (2006), Pham et al. (2003), Habib and Ljungqvist (2001), Brennan and Franks (1997), and Booth and Chua (1996) show that underpricing is an effective substitute for promotional activities, suggesting a tradeoff between underpricing and promotional activities in that underpricing may help reduce the cost of promotional activities.

We also hypothesize that hiring reputable underwriters augments the chance of post-IPO firm sale given their large capital capacity to provide aftermarket price stability support, close and repeated relationship with certain investors to prevent flipping as well as broader network with potential acquirers. Further, Chemmanur and Krishnan (2012) show that underwriters with better reputations set the equity price higher compared to underwriters with less reputation.[7] Hence, we expect that likely bidders – for whom capital infusion, information production, secondary market liquidity and control retention are relatively more important than promotional mechanisms of IPOs – are more likely to hire relatively less reputable underwriters. Potential target firms are more likely to emphasize promotional mechanisms and hire reputable underwriters to augment the chance that the firm will be acquired shortly and receive a high valuation.

Defining the marketing of a security “as actions that influence the demand and hence affect the price of the security without necessarily discovering any private information on the intrinsic value of the security,” Huang and Zhang (2011) and Dong et al. (2011) consider such activities as marketing function of IPO design: (i) any promotional activities in the road shows; (ii) aftermarket price support-price stability (aftermarket purchases, market making, and penalty bids); (iii) analyst coverage. We posit that these activities are likely to be emphasized in IPOs of likely targets.

2.6. Market conditions and growth strategies

IPOs volume often clusters in time in so-called “hot” IPO markets, and firms may avoid issuing in periods where few other good quality firms issue (Choe et al., 1993; Subrahmanyam and Titman, 1999). Clustered volume in IPOs also appears related to valuation cycles in equity markets (Schultz, 2003). Consequently, market timing and strategic waiting – in addition to operational demand for funds – may explain fluctuations in IPO volume over time (Baker and Wurgler, 2002; Çolak and Günay, 2011). We investigate the phenomenon of market timing from the perspectives of both M&A market and equity market by including several proxies that track M&A activity and equity valuation. We investigate if eventual bidders or targets go public based on opportunistic equity market conditions or M&A market conditions.

Alimov and Mikkelson (2012) find that firms that spend, on average, more in post-IPO acquisitions tend to be firms going public during the periods of favorable investor sentiments. Emphasizing the importance of timing IPOs, Çolak and Günay (2011) show that some IPO firms strategically delay their issuance. Newly public firms which become targets shortly after IPO may also be willing to time the market, but they may be more constrained by the pressure of achieving economies of scale, speeding products to market, and satisfying the desires for exit by pre-IPO investors. Poulsen and Stegemoller (2008) find that compared to private firms subject to takeovers (sellouts), firms which go public tend to have more growth opportunities and need for capital to fuel their growth. But, even for publicly-held firms, Gao et al. (2013) argue that slow organic growth as an independent company is less attractive than quickly achieving economies of scale and scope via mergers in a competitive “eat or be eaten” environment. This is especially true for newly public firms with no acquisition incentives or growth prospects or those which underperform experiencing deteriorations in business competitiveness. Jain and Kini (1999) find that smaller firms tend to be acquired or fail and are thus less likely to stay independent after IPO. On the other hand, experienced acquirers may perpetuate their acquisition

[5] Price stability support is not directly observable. The proxy for price stability support that we employ is, in general, in line with Eisenstadt and Li (2005). In an unreported table, we additionally used two more proxies: percentage of overallotment option exercised, and percentage of negative initial returns, as discussed by Eisenstadt and Li (2005). Generally, the proxies provide consistent results.

[6] Substantial amount of marketing and promotional effort during the IPO may eliminate or at least mitigate the necessity of publicity during the selling process.

[7] Chemmanur and Krishnan (2012) argue that higher reputation underwriters are able to attract a greater number of quality market participants, yielding higher IPO valuations by increasing heterogeneity in investor beliefs and tightness of short-sale constraints.

agenda by opportunistically engaging in a series of merger deals. Consequently, likely targets may appear to be less likely to behave opportunistically with respect to the timing of an IPO than acquisition-motivated likely bidders. We measure market timing by (a) the number of acquisitions in the quarter prior to the IPO, divided by total number of firms in the revised 49 Fama-French industries, a measure reflecting industry-wide M&A intensity, and (b) Baker-Wurgler sentiment index in the quarter to prior to the IPO. The former tracks the movement of the M&A wave in the market, while the latter proxies mostly equity market conditions.

2.7. Control variables

We control for other variables predictive of M&A activity, such as venture capital backing, age, market-to-book, ROA, book leverage, and year and industry fixed effects. For example, venture capital backing, among these variables, is expected to be positively associated with the likelihood of becoming a target because venture capital investors with typically limited investment horizons are more likely to consider the IPO as part of a staged sale of their pre-IPO stakes.

3. Predicting post-IPO merger activity

We investigate how public information observed at the time of the IPO affects the likelihood and timing of subsequent M&A activity via several alternative estimation techniques. Our initial reported results are based on logit models which estimate the likelihood that a newly public firm becomes a bidder or target, respectively, as a function of IPO-characteristics and other variables. In the same table, we also report additional results using a multinomial logit model with several alternative post-IPO outcomes. In addition, we report results from a hazard model that estimates the hazard from IPO until a firm becomes a bidder or target. Hazard models enable us not to rely on a specific time window in which a merger activity is observed, and therefore, provide an important robustness check. In proportional hazard models, the coefficients are exponentiated and reported as odd ratios, and the signs of the coefficients are interpretable similar to those in the other regressions in that a coefficient >1 implies an increased hazard for M&A activity.

In all of the estimations reported in this study, we acknowledge that we do not eliminate sample selection bias with respect to private firms choosing to go public at a point in time. Entirely eliminating this selection bias would require us to observe the entire universe of private firms before they go public. Trying our best to distinguish future bidders and targets based on what we can observe, we, like investors, can only work with observable data of firms that do eventually go public.[8] Further, we put ourselves in the shoes of investors at the time of an IPO in trying to predict and act upon whether newly public firms will subsequently engage in M&A as bidders or targets. Consistent with Bharath and Dittmar (2010) or “early warning models” in established bankruptcy/bank failure literature, prediction models, by definition, are not concerned with direction of causality or endogeneity. Thus, we center our focus on associations which yield the strongest predictive power in explaining post-IPO M&A activity.

3.1. Identifying likely bidders

Specifications (1) to (3) in Table 4 estimate logit models where the dependent variable is the log likelihood that a newly public firm becomes a bidder firm within 1 year to 3 years of its IPO date. Since the IPO-wave during the bubble period has distinct features, specification (4) excludes bubble period and re-estimates the likelihood of making acquisitions within 1 year of the IPO date. We also extend our analysis to two alternative forms of estimation for post-IPO M&A activity. The first is a multinomial logit model which predicts the likelihood of becoming a target or a bidder with a base case of not engaging in M&A activity, as shown in specification (5) to (7). Finally, column (8) estimates a hazard model for the time that elapses between an IPO and the firm becoming a bidder.

The results show that newly public firms appear more likely to become bidders when underpricing is more severe, which is consistent with likely bidders' preference for a wide range of investors and secondary-market liquidity. This finding is robust across different estimation models and time-windows. A one-standard deviation increase in underpricing across different specifications is, on average, associated with a 10.2% increase in the likelihood of post-IPO acquisitions among the newly public firms. The amount and structure of proceeds, in general, appear to affect the likelihood that a newly public firm becomes a bidder. The coefficients on the logged values of primary proceeds are mostly positive and significant, suggesting that the more dollar amount of proceeds obtained during the IPO, the higher chance of engaging in acquisitions. *Disclosure Dummy* and *Post-IPO 20-day Return* variables provide some evidence that IPO serves somewhat as valuable channels for information generation between firm managers and market participants. While firms which reveal their intention of engaging in post-IPO acquisitions at the time of IPO are more likely to become bidders, firms which receive more positive reactions from the market during the first 20 trading days of their shares are also more likely to involve in an acquisition within the first year of their IPOs. The role of market feedback mechanism in implementing post-IPO acquisitions seems to decay for the following years, as *Post-IPO 20-day Return* becomes insignificant in the specifications of two to three years post IPO.

[8] The choice and timing of using private or public capital markets in going public or privatization depend upon numerous institutional and firm-specific factors (Megginson et al., 2004). In unreported estimations, we attempted to eliminate the sample selection bias by working with the universe of Compustat firms. In those estimations, we did not obtain a statistically significant Inverse Mills ratio, implying an absence of selection bias. Nevertheless, we acknowledge that Compustat universe mostly includes large private and public firms, and does not allow us to observe small private firms for some time before they go public.

Institutional ownership variables measured within one quarter following the IPO indicate that firms that allocate shares to institutional investors are more likely to engage in acquisitions. In untabulated specifications, we find that this result is driven by transient institutional investors, suggesting that a firm that acquires shortly after its IPOs prefers allocation of shares to such institutions. *IO Breadth* consistently carries negative coefficients, yet the finding is rather weak. Firms issuing dual class shares are less likely to engage in future acquisitions. Dual class shares serve as a strict control instrument, while stock-financed acquisitions dilute pre-IPO shareholders' influence on the firm. Likely bidders which intend to generate acquisition currency and swap the firm stock to facilitate acquisitions probably refrain from a dual class share structure.

The variables related to underwriting and promotional activities appear to have little ability to distinguish likely bidders from other newly public firms in general. In four out of eight instances, *All-star Analyst Dummy* receives negative and significant coefficients, suggesting that IPO issuers that emphasize analyst coverage in their IPOs are less likely to become bidders. This result is strong and robust across different econometric methods for likely bidders that rapidly engage in acquisitions within *one year of IPO*. This might be due to opportunistic market timing behavior. Not necessarily interested in generating publicity or selling shares with greater investor recognition (which slowly develops over time) as a result of reputable analyst coverage, insiders at acquisition-motivated firms may rather prefer keeping IPO related costs low and opportunistically settling acquisitions in a short period of time.[9] Rather, all-star analyst coverage and reputable underwriters are expected to play a considerable role for likely targets, as discussed earlier.

Recent equity market sentiment and/or industry-wide merger intensity can affect the motivation and the likelihood of a newly public firm engaging in M&A as a bidder. *Pre-IPO sentiment*, a proxy for optimism about the equity market, does not seem to have an impact on the likelihood of becoming a bidder, yet industry acquisition intensity is positively associated with the likelihood of becoming a bidder. In other words, likely bidders do not time the IPO based on the sentiment of equity markets, but rather time their IPOs based on the wave of M&A market.[10] The marginal effect is not trivial. A one-standard deviation increase in *Industry Acquisition Intensity* across different specifications is, on average, associated with a 4.2% increase in the probability of post-IPO acquisitions. IPO firms that rely on organic growth are less likely to become bidders, but those that have undergone at least one merger deal prior to IPO are statistically more prone to become bidders. Among control variables, we find that more profitable and leveraged firms have a higher chance of engaging in post-IPO acquisitions. All specifications include industry fixed effects and year fixed effects which might capture underlying unobservable factors.

A natural question is whether the IPO-specific variables we include jointly predict bidder likelihood with any economic significance. To assess the importance of IPO characteristics, we calculate the predicted likelihood of becoming a bidder using three alternative specifications from the semi-parametric Cox hazard model. The first model specification, which we refer to as the prediction only with disclosure dummy (whether the issuer discloses future acquisitions in the section of use of proceeds in the prospectus), includes only industry-specific terms, calendar year intercept terms, and disclosure dummy. This is a primitive "naive" model for predicting bidder likelihood. The second model specification, which we refer to as the prediction without deal structure, also includes industry-specific intercept terms, calendar year intercept terms, venture capital backing dummy, firm age as well as financial statement ratios. The third model adds all the other IPO-specific variables to estimate an enhanced probability of becoming a bidder within three years post IPO, which we refer to as the prediction with deal structure. Fig. 2 displays the frequency of actual bidders within 3 years of IPOs across the sample after being segmented into estimated probability quartiles by each of the three models. The enhanced model which includes IPO-specific terms appears to materially improve on the performance of the model. In particular, the frequency of becoming a bidder in the lowest quartile of probability implied by the enhanced model is 24.9%, compared to 35.4% and 29.9% implied by the naïve model and the model without deal structure variables, respectively. Commensurately, the frequency of bidders in the highest quartile of probability implied by the enhanced model is 66.0%, compared to 57.5% and 60.1% obtained from the naïve model and the model without deal structure variables, respectively.

3.2. Identifying likely targets

Specifications (1) to (4) in Table 5 report estimated logit models where the dependent variable is the log likelihood that a newly public firm becomes a target firm within a few years of its IPO date. Table 5 is structured similarly to Table 4, and we focus our subsequent discussion on the IPO-specific characteristics and their effects on target likelihood.

As opposed to the results concerning bidders, pricing, proceeds and information production mechanisms of IPO deal structure play a minor role in distinguishing the likelihood of becoming targets. However, we observe that share overhang is significantly associated with the probability of an IPO firm becoming a post-IPO target. This is consistent with the literature which finds that shares retained by pre-IPO shareholders are mostly due to the lockup period. While pre-IPO shareholders cannot sell due to the lockup agreement, they may be motivated to engage the IPO firm in a firm sellout in an attempt to cash out.

There seem to be distinct share allocation strategies among IPO firms which subsequently become post-IPO bidders versus targets. The coefficient on *IO Ratio* is significant and much larger than that observed in Table 4. In untabulated follow-up tests, we

[9] Cliff and Denis (2006) argue that providing analyst coverage recommendations might be costly. The cost can be materialized directly as cash or indirectly as further/deeper underpricing.

[10] We first make sure that the effects of *Industry Acquisition Intensity* are not due to multi-collinearity. In addition to Baker-Wurgler sentiment index, we alternatively use stock returns of S&P index and stock turnover as additional measures to capture the equity market conditions. In all of these specifications, *Industry Acquisition Intensity* preserves the statistical significance.

Table 4
Predicting post-IPO acquirers.

	(1) Logit model	(2)	(3)	(4)	(5) Multinomial logit model	(6)	(7)	(8) Proportional hazard model
	IPO + 1 yr	IPO + 2 yrs	IPO + 3 yrs	IPO + 1 yr (excluding internet bubble period)	IPO + 1 yr	IPO + 2 yrs	IPO + 3 yrs	
Pricing & proceeds								
Underpricing	0.0085***	0.0056***	0.0050***	0.0094***	0.0085***	0.0052***	0.0043**	1.0029**
	(0.0000)	(0.0025)	(0.0071)	(0.0003)	(0.0000)	(0.0061)	(0.0259)	(0.0197)
Log(Primary Proceeds)	0.1830*	0.1948**	0.1955**	0.1822*	0.1877*	0.2127**	0.2273**	1.0541
	(0.0686)	(0.0249)	(0.0184)	(0.0805)	(0.0639)	(0.0195)	(0.0111)	(0.2636)
Log(Secondary Proceeds)	0.0085	0.0577	0.0369	0.0246	0.0119	0.0626	0.0478	1.0056
	(0.8435)	(0.1470)	(0.3428)	(0.5865)	(0.7828)	(0.1223)	(0.2393)	(0.8132)
Log(Overallotment Proceeds)	−0.1070	−0.0412	−0.0144	−0.1460	−0.1197	−0.0569	−0.0273	1.0564
	(0.3498)	(0.6859)	(0.8837)	(0.2288)	(0.3062)	(0.5941)	(0.7958)	(0.3438)
Price Revision	−0.0015	0.0027	0.0018	−0.0004	−0.0012	0.0032	0.0021	1.0000
	(0.5955)	(0.3146)	(0.4888)	(0.8949)	(0.6981)	(0.2522)	(0.4579)	(0.9793)
Disclosure Dummy	0.4339***	0.6550***	0.5200***	0.4516**	0.4504***	0.6830***	0.5977***	1.3226***
	(0.0054)	(0.0000)	(0.0007)	(0.0117)	(0.0045)	(0.0000)	(0.0004)	(0.0028)
Post-IPO 20-day Return	0.4850**	0.3448	0.3169	0.3280	0.4527*	0.2886	0.2379	1.2461*
	(0.0370)	(0.1165)	(0.1473)	(0.2421)	(0.0578)	(0.2071)	(0.3017)	(0.0929)
Ownership structure								
Overhang	0.0095	−0.0023	−0.0051	0.0173	0.0151	0.0036	0.0044	1.0044
	(0.6871)	(0.9118)	(0.7979)	(0.4871)	(0.5196)	(0.8667)	(0.8351)	(0.6669)
IO Ratio	1.1781***	0.9516**	0.9175**	1.3475***	1.2539***	1.1391***	1.1939***	1.5977*
	(0.0092)	(0.0245)	(0.0273)	(0.0055)	(0.0044)	(0.0069)	(0.0048)	(0.0686)
IO Breadth	−0.0074	−0.0108**	−0.0112**	−0.0056	−0.0076	−0.0110**	−0.0117**	0.9945*
	(0.1934)	(0.0385)	(0.0280)	(0.3846)	(0.1552)	(0.0307)	(0.0211)	(0.0819)
Institutional Ownership HHI	−0.3147	−0.2290	−0.2551	−0.3900	−0.3082	−0.1643	−0.1823	0.7695*
	(0.2421)	(0.3205)	(0.2412)	(0.1808)	(0.2457)	(0.4910)	(0.4263)	(0.0676)
Dual Class Dummy	−0.4516**	−0.4516**	−0.3407*	−0.3279	−0.4759**	−0.5701***	−0.4791***	0.8618
	(0.0255)	(0.0135)	(0.0550)	(0.1084)	(0.0176)	(0.0022)	(0.0081)	(0.1729)
Promotion & underwriting								
All-star Analyst Dummy	−0.3521**	−0.1845	−0.2030	−0.3712**	−0.3585**	−0.1775	−0.2029	0.8480**
	(0.0127)	(0.1637)	(0.1222)	(0.0182)	(0.0142)	(0.1971)	(0.1446)	(0.0462)
Bid-ask Spread Growth	−0.0006	0.0011	0.0008	−0.0000	−0.0005	0.0017	0.0010	0.9999
	(0.6978)	(0.4384)	(0.5622)	(0.9877)	(0.7230)	(0.2394)	(0.4662)	(0.8981)
Bookrunner Rank	0.0132	0.0253	0.0131	0.0028	0.0173	0.0321	0.0248	1.0180
	(0.6639)	(0.3586)	(0.6248)	(0.9297)	(0.5826)	(0.2694)	(0.3807)	(0.2722)
Market condition & growth strategy								
Pre-IPO Sentiment	−0.0663	−0.2478	−0.1363	−0.1842	−0.0482	−0.2798	−0.2163	0.8385**
	(0.8525)	(0.4532)	(0.6708)	(0.6359)	(0.8957)	(0.4170)	(0.5254)	(0.0315)
Industry Acquisition Intensity	3.4713***	2.3965***	1.6535*	3.4897***	3.4655***	2.6140***	1.7527*	5.4864***
	(0.0002)	(0.0070)	(0.0568)	(0.0006)	(0.0003)	(0.0045)	(0.0563)	(0.0004)
Organic Growth/Assets	−2.2902***	−2.7674***	−1.9633***	−2.1357***	−2.3114***	−2.8834***	−2.1816***	0.3125***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Pre-IPO Acquisition Dummy	0.8607***	1.0853***	1.0761***	1.0665***	0.8767***	1.1346***	1.1037***	1.7310***
	(0.0006)	(0.0000)	(0.0001)	(0.0001)	(0.0002)	(0.0000)	(0.0001)	(0.0002)
Control								
VC Dummy	0.1306	0.0640	0.1671*	0.0633	0.1397	0.1190	0.2626***	1.0927
	(0.2180)	(0.5107)	(0.0767)	(0.5733)	(0.1865)	(0.2288)	(0.0077)	(0.1224)
Age	0.0007	0.0003	0.0006	0.0001	0.0004	−0.0006	−0.0006	1.0002
	(0.8035)	(0.9239)	(0.8310)	(0.9823)	(0.8947)	(0.8249)	(0.8156)	(0.9071)
Market-to-Book	−0.0021	0.0370*	0.0320*	−0.0411	−0.0030	0.0313	0.0386*	1.0122
	(0.9155)	(0.0507)	(0.0889)	(0.1798)	(0.8795)	(0.1023)	(0.0505)	(0.2685)
ROA	0.5662**	0.5524**	0.8346***	0.5712*	0.5479**	0.5882**	0.9413***	1.9480***
	(0.0419)	(0.0298)	(0.0006)	(0.0712)	(0.0453)	(0.0181)	(0.0001)	(0.0000)
Book Leverage	0.9512***	0.8167***	0.5062**	1.0578***	0.9844***	0.8607***	0.5423**	1.0824
	(0.0001)	(0.0004)	(0.0253)	(0.0000)	(0.0001)	(0.0003)	(0.0232)	(0.5753)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Observations	2971	2979	2979	2712	2993	2993	2993	2993
Pseudo R-squared	0.110	0.103	0.0886	0.110	0.124	0.115	0.105	

find that compared to likely bidders which attract transient institutional investors for the post-IPO ownership structure, likely targets lean toward dedicated institutional investors and prefer more concentrated institutional ownership structures which can facilitate firm-sale over a longer period, i.e., within two to three years of their IPOs. Overall, these results imply that likely bidders tend to avoid share allocations to institutional investors and/or prefer transient institutional investors. Likely targets, on

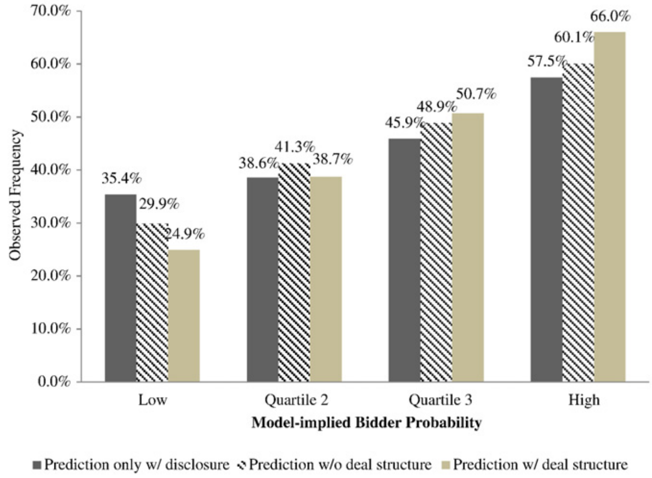

Fig. 2. Observed frequency of bidder activity within 3 years of IPO and model-implied bidder probability. This figure displays the frequency of actual bidders across the sample after being segmented into estimated probability quartiles by each of the three models. The first model specification, which we refer to as the prediction only with disclosure (whether the issuer discloses future acquisitions in the section of use of proceeds in the prospectus), includes only industry-specific terms, calendar year intercept terms, and disclosure dummy. The second model, which we refer to as the prediction without deal structure, includes industry-specific intercept terms, calendar year intercept terms, venture capital backing dummy, firm age as well as financial statement ratios. The third model adds all the other IPO-specific variables to estimate an enhanced probability of becoming a bidder, which we refer to as the prediction with deal structure.

the other hand, tend to attract more concentrated and more dedicated institutional investors that may be active in monitoring firms. These institutional investors might increase transparency, reduce asymmetric information, and/or play a role in facilitating firm sale to potential acquirers.

Some of the variables related to underwriting and promotional mechanisms predict target likelihood. In particular, underwriter quality is positively related to target likelihood, and size of post-IPO bid-ask spread growth observed over the 2 months after the IPO is also positively related to target likelihood. This is consistent with likely targets desiring post-IPO price support by their reputable underwriters, since price support has been shown to lead to lower post-IPO spreads in the initial weeks after the IPO.

Neither equity market sentiment nor industry M&A intensity in pre-IPO period has any statistical association with the likelihood of becoming a target, suggesting that the motivation for going public is not related to timing the market for selling out the firm opportunistically. Similar to the results for likely bidders, organic-growth-oriented IPO issuers which build up their R&D and capital expenditures are less likely to become targets. Also conforming to our expectations, IPOs backed by venture capitalists are more likely to become targets soon after IPO.

Fig. 3 shows the how predicting likely target firms is enhanced by the inclusion of IPO-specific variables in the Cox Hazard model from Table 5. Relative to the naïve model based on only disclosure dummy and the model without deal structure based solely on venture capital backing, firm age, financial ratios, industry fixed effects, and year fixed effects, the enhanced model (the prediction with deal structure) more clearly distinguishes likely targets from unlikely targets within three years post IPO. In particular, for the quartile of firms with the lowest predicted likelihood of becoming a target, the enhanced model identifies targets at a 7.1% frequency compared to 10.2% for the prediction model only with disclosure dummy and 9.0% for the prediction model without deal structure variables. At the other end of the spectrum, the enhanced model better identifies likely targets in the highest probability quartile, moving the actual frequency of observed target frequency from 21.1% and 22.4% to 24.4%.

Overall, the results reported in Tables 4 and 5 jointly suggest that there are stark differences in IPO deal structures of acquisition-motivated and firm-sale motivated IPO issuers. First, although there is no evidence that likely targets time their IPOs, the findings support the view that likely bidders do opportunistically time their public offerings. Second, the underpricing and IPO proceeds are strong predictors of post-IPO acquisitions. Post-IPO bidders appear to underprice to a greater extent and generate larger proceeds at the time of IPOs, while post-IPO targets do not. In addition, the results provide new evidence to the view that the underwriters of likely targets tend to engage in price stability activities after the IPO, and firms with larger share

Notes to Table 4:

This table presents several specifications estimating the determinants of probability of post-IPO bidders. Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008, and we track IPO firms' acquisition activity for up to three years post-IPO. Specifications (1) to (3) report estimated logit models where the dependent variable is the log likelihood that a newly public firm becomes a bidder firm within 2 year to 3 years of its IPO date. Specification (4) excludes bubble period and re-estimates the likelihood of making acquisitions within 1 year of the IPO date. Analyses are also extended to two alternative forms of estimation for post-IPO M&A activity. Specifications (5) to (7) employ multinomial logit models that predict the likelihood of becoming a target or a bidder with a base case of not engaging in any M&A activity and report the results only related to bidder likelihood. Finally, column (8) estimates a hazard model for the time that elapses between an IPO and the firm becoming a bidder firm. Each estimation includes a set of control variables such as *VC back dummy, firm age, market-to-book, ROA, book leverage*, and firm fixed effects and industry fixed effects based on the revised FF 49 industry classification. The coefficients in specifications (1) to (7) reflect marginal effects, and the coefficients in specification (8) are exponentiated and reported as risk ratios. P-values are reported in parentheses and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively. Proceeds and assets are expressed in 2003 dollars. All IPO variables are defined in Appendix A.

Table 5
Predicting post-IPO targets.

	(1) Logit model	(2)	(3)	(4)	(5) Multinomial logit model	(6)	(7)	(8) Proportional hazard model
	IPO + 1 yr	IPO + 2 yrs	IPO + 3 yrs	IPO + 1 yr (excluding internet bubble period)	IPO + 1 yr	IPO + 2 yrs	IPO + 3 yrs	
Pricing & proceeds								
Underpricing	−0.0002	−0.0013	−0.0010	0.0046	−0.0002	−0.0036	−0.0031	1.0004
	(0.9630)	(0.6256)	(0.6710)	(0.4253)	(0.9790)	(0.4016)	(0.3826)	(0.7233)
Log(Primary Proceeds)	0.0858	0.0484	0.0763	0.1975	0.1972	0.1898	0.2494	1.0475
	(0.6975)	(0.7466)	(0.5256)	(0.4833)	(0.5942)	(0.3563)	(0.1556)	(0.4509)
Log(Secondary Proceeds)	0.0050	−0.0199	0.0681	−0.0260	0.1045	0.0567	0.0691	1.0529*
	(0.9649)	(0.7657)	(0.1877)	(0.8489)	(0.4756)	(0.5229)	(0.3485)	(0.0627)
Log(Overallotment Proceeds)	−0.1282	0.0666	0.1954	−0.2023	−0.4918	−0.1657	−0.0835	1.0288
	(0.6542)	(0.7041)	(0.1616)	(0.5668)	(0.2020)	(0.4724)	(0.6686)	(0.7006)
Price Revision	0.0063	−0.0021	−0.0022	0.0104	0.0102	0.0028	−0.0008	1.0008
	(0.4243)	(0.6338)	(0.5598)	(0.2723)	(0.3026)	(0.6418)	(0.8678)	(0.6899)
Disclosure Dummy	0.5040	−0.0049	0.1952	0.5848	0.4811	0.1778	0.3950	0.8943
	(0.1983)	(0.9841)	(0.3252)	(0.2811)	(0.3615)	(0.6127)	(0.1870)	(0.3584)
Post-IPO 20-day Return	−1.1113*	−0.2180	0.0985	−1.2995	−0.9420	−0.5181	−0.4851	0.9681
	(0.0927)	(0.5689)	(0.7425)	(0.1668)	(0.2748)	(0.2964)	(0.2447)	(0.8362)
Ownership structure								
Overhang	0.1291***	0.0684***	0.0658***	0.1144**	0.1339**	0.0613	0.0559*	1.0196*
	(0.0008)	(0.0064)	(0.0022)	(0.0141)	(0.0179)	(0.1561)	(0.0977)	(0.0871)
IO Ratio	1.6266	2.0810***	1.4488***	2.5945*	3.0536*	2.1413**	1.9182**	2.1804***
	(0.1691)	(0.0013)	(0.0053)	(0.0675)	(0.0532)	(0.0168)	(0.0120)	(0.0069)
IO Breadth	−0.0033	−0.0002	−0.0053	−0.0171	−0.0038	0.0009	−0.0023	0.9990
	(0.8136)	(0.9831)	(0.3761)	(0.4677)	(0.8510)	(0.9309)	(0.7973)	(0.7706)
Institutional Ownership HHI	0.5090	0.8022*	0.6948**	1.1383	0.3988	1.0708**	0.7988*	1.1832
	(0.5383)	(0.0647)	(0.0434)	(0.1997)	(0.7165)	(0.0461)	(0.0864)	(0.3418)
Dual Class Dummy	−1.3394	−1.4125***	−1.0375***	−1.9201	−1.1999	−1.8264***	−1.1163***	0.5680***
	(0.1269)	(0.0015)	(0.0014)	(0.1016)	(0.1661)	(0.0043)	(0.0086)	(0.0003)
Promotion & underwriting								
All-star Analyst Dummy	0.0709	0.0910	0.1411	−0.0242	−0.1989	0.0463	−0.0415	1.1317
	(0.8474)	(0.6650)	(0.3982)	(0.9584)	(0.6873)	(0.8662)	(0.8591)	(0.1838)
Bid-ask Spread Growth	0.0024	0.0043**	0.0017	0.0008	0.0015	0.0053*	0.0009	1.0016*
	(0.5055)	(0.0466)	(0.3689)	(0.8766)	(0.7684)	(0.0640)	(0.7428)	(0.0846)
Bookrunner Rank	0.1322	0.0989*	0.0950**	0.1860	0.2003	0.1081	0.1160**	1.0716***
	(0.2281)	(0.0765)	(0.0258)	(0.1578)	(0.1105)	(0.1221)	(0.0493)	(0.0008)
Market condition & growth strategy								
Pre-IPO Sentiment	1.0071	−0.3316	−0.1141	1.0660	0.8557	−0.3840	−0.5221	0.8430*
	(0.4293)	(0.5711)	(0.8081)	(0.4281)	(0.5497)	(0.5977)	(0.4196)	(0.0911)
Industry Acquisition Intensity	−0.1762	1.2301	0.1266	2.6129	−0.7868	1.7946	0.0379	1.8834
	(0.9522)	(0.3976)	(0.9112)	(0.3812)	(0.8399)	(0.3685)	(0.9822)	(0.2428)
Organic Growth/Assets	−2.4211*	−1.3417**	−1.3638**	−3.3355**	−0.5958	−1.1937	−1.3926*	1.1660
	(0.0770)	(0.0489)	(0.0129)	(0.0302)	(0.6874)	(0.1629)	(0.0519)	(0.5632)
Pre-IPO Acquisition Dummy	0.0733	0.1813	−0.4773	0.1928	0.4471	0.3885	0.1396	0.9548
	(0.9052)	(0.6037)	(0.1566)	(0.7879)	(0.5909)	(0.5052)	(0.7946)	(0.7903)
Controls								
VC Dummy	0.6072*	0.3702**	0.2849**	0.7496**	0.3409	0.5709***	0.6112***	1.3427***
	(0.0583)	(0.0234)	(0.0303)	(0.0427)	(0.3564)	(0.0086)	(0.0009)	(0.0000)
Age	−0.0082	−0.0088	−0.0061	−0.0006	−0.0207	−0.0113	−0.0105*	0.9981
	(0.5114)	(0.1007)	(0.1451)	(0.9574)	(0.2035)	(0.1280)	(0.0933)	(0.3718)
Market-to-Book	0.0293	−0.0460	−0.0229	0.0414	−0.0170	−0.0529	0.0265	0.9934
	(0.4979)	(0.1406)	(0.3867)	(0.5269)	(0.7772)	(0.2317)	(0.4289)	(0.6176)
ROA	-0.3086	0.1886	0.3365	0.1425	−0.3635	0.6385	0.8640*	1.6503***
	(0.6387)	(0.6533)	(0.3165)	(0.8643)	(0.6705)	(0.2559)	(0.0654)	(0.0036)
Book Leverage	0.6700	0.4965	0.1782	−0.4666	1.1232	0.5118	0.2161	0.6084***
	(0.3952)	(0.1871)	(0.5748)	(0.5967)	(0.2105)	(0.3358)	(0.6402)	(0.0074)
Industry fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	No
Observations	2422	2866	2957	2205	2993	2993	2993	2993
Pseudo R-squared	0.101	0.0827	0.0788	0.117	0.124	0.116	0.106	

overhang are more likely to become targets. Third, likely targets tend to rely on reputable underwriters during IPO process. Finally, positive market feedback for IPO firms with acquisition intentions plays a facilitating role in post-IPO acquisitions. The result is strong for the first year of IPO, but not robust for acquisitions that take place two to three years post IPO. It might be that the role of market feedback mechanism in implementing the post-IPO acquisitions decays over time. On the other hand, a larger positive

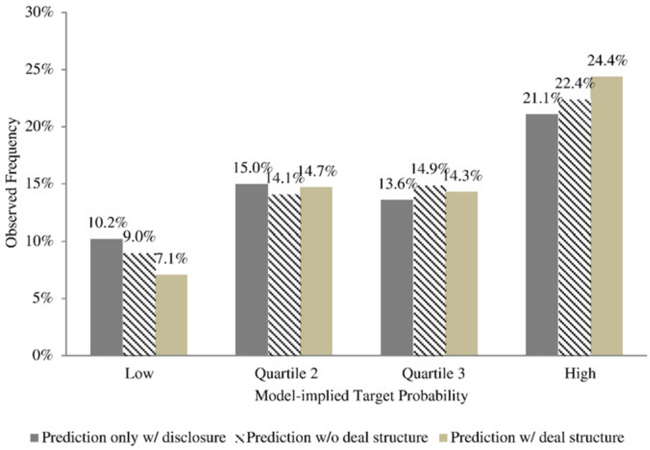

Fig. 3. Observed frequency of target activity within 3 years of IPOs and model-implied target probability. This figure displays the frequency of actual targets across the sample after being segmented into estimated probability quartiles by each of the three models. The first model specification, which we refer to as the prediction only with disclosure (whether the issuer discloses future acquisitions in the section of use of proceeds in the prospectus), includes only industry-specific terms, calendar year intercept terms, and disclosure dummy. The second model, which we refer to as the prediction without deal structure, includes industry-specific intercept terms, calendar year intercept terms, venture capital backing dummy, firm age as well as financial statement ratios. The third model adds all the other IPO-specific variables to estimate an enhanced probability of becoming a bidder, which we refer to as the prediction with deal structure.

reaction from the market during the first 20 trading days is either not related or negatively related with becoming a takeover target. This suggests that poor performers or firms with not much of a growth prospect, reflected in investor valuations during the IPO process, are more likely to become targets.

The smoothed hazard functions depicted in Fig. 4 present market timing motives between likely bidders and targets. The baseline hazard is evaluated at the mean values of the independent variables in the last columns of Tables 4 and 5, and smoothed using nonparametric kernel density estimator consistent with Allison (1995). Clearly, market sentiment has a substantially different impact on the likelihood of post-IPO M&A activities. First, hazard rates are larger by about an order of magnitude for likely-bidder-IPO firms, suggesting that acquisition tendencies are substantially larger than firm-sale tendencies among newly public firms. Second, even though both likely bidders and targets have a higher chance to engage in an M&A deal during a hot M&A market, likely bidders appear to exploit the hot market to a larger extent. At the maximum points of the hazard functions, the likelihood of making acquisitions for newly public firms in a hot market is double that in normal market. The likelihood of selling out a firm in a hot market, however, is only about one-third higher than that in a normal market. Third, compared to likely targets, likely bidders that tend to take advantage of a hot market act considerably sooner after their IPOs. As seen in the top figure, the break-even point at which three hazard functions of acquisitions for hot, normal, and cold markets intersect occurs around 1000 days after the IPO dates. The break-even point for likely targets is around 2200 days after the IPO dates. Fourth, while we see a monotonic relationship between the likelihood of acquisitions/firm sales and M&A market intensity up to the break-even point, the hazard of making an acquisition is, on average, not distinguishable in a hot market time from that in a normal or cold market time after the break-even point. Interestingly, however, the hazard of selling out is substantially higher during a cold market time for a long time after the break-even point. This suggests that a considerable portion of post-IPO firm-sales are driven by necessity, such as underperformance, financial difficulty or deterioration in the competitiveness of the firms, consistent with Gao et al. (2013). In a nutshell, from the smoothed hazard functions and regression analyses, we conclude that market timing plays a particularly important role in the decisions of going public of likely bidders, and that likely bidders, on average, time their IPOs based on the M&A market rather than the equity market conditions.

Notes to Table 5:

This table presents several specifications estimating the determinants of probability of post-IPO targets. Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008, and we track IPO firms' acquisition activity for up to three years post-IPO. Specifications (1) to (3) report estimated logit models where the dependent variable is the log likelihood that a newly public firm becomes a target firm within 1 year to 3 years of its IPO date. Specification (4) excludes bubble period and re-estimates the likelihood of making acquisitions within 1 year of the IPO date. Analyses are also extended to two alternative forms of estimation for post-IPO M&A activity. Specifications (5) to (7) employ multinomial logit models that predict the likelihood of becoming a target or a bidder with a base case of not engaging in any M&A activity and report the results only related to target likelihood. Finally, column (8) estimates a hazard model for the time that elapses between an IPO and the firm becoming a target firm. Each estimation includes a set of control variables such as *VC back dummy, firm age, market-to-book, ROA, book leverage*, and firm fixed effects and industry fixed effects based on the revised FF 49 industry classification. The coefficients in specifications (1) to (7) reflect marginal effects, and the coefficients in specification (8) are exponentiated and reported as risk ratios. *P*-values are reported in parentheses, and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively. Proceeds and assets are expressed in 2003 dollars. All IPO variables are defined in Appendix A.

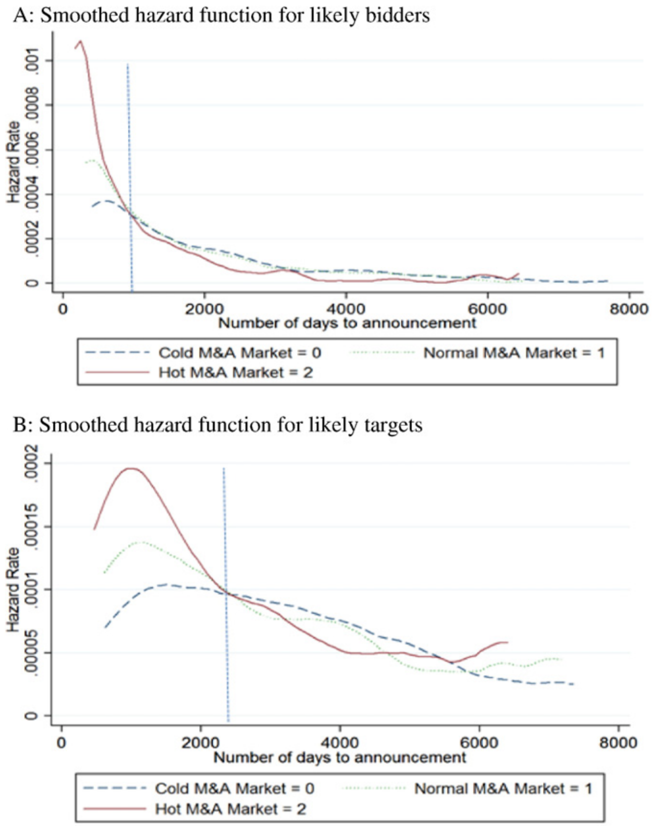

Fig. 4. Smoothed hazard functions. These figures depict the smoothed hazard functions of likely bidders and targets. The baseline hazard is evaluated at the mean values of the independent variables in the last column of Tables 4 and 5, and smoothed using nonparametric kernel density estimator consistent with Allison (1995).

3.3. ROC curve analysis

We also test the power of our prediction models using a Receiver Operating Characteristic (ROC) curve analysis. A ROC curve depicts the proportion of accurate prediction rate, known as sensitivity, against false alarms ("1-specificity"), generated by a forecasting model. Hence, a ROC curves indicates a trade-off between sensitivity and specificity about predicting M&A activities of post-IPO firms. In the context of our study, Y-axis (sensitivity) shown in Fig. 5a and b represent the proportion of acquisitions and sell-outs accurately detected by our model, while X-axis, 1-specificity, represents inaccurate, false alarms generated by the model. Arbitrary forecasts about the M&A activities of post IPO firms would be expected to lie on the diagonal line. The accuracy of our prediction models is shown in the area under the curves.

The findings from the ROC curves provide evidence that the model based on IPO characteristics of likely bidders and targets can substantially detect their post-IPO motives. The area under the curve in Fig. 5a and b indicates that our baseline Cox hazard model can predict which IPO firms will engage in acquisitions or sell-out within 3 years of their IPOs 71.3% and 70.0% of the time, respectively. These findings suggest that inherent characteristics observed right at the very beginning stage of a firm's public life provide valuable signals about not only firms' decisions to go private (Bharath and Dittmar, 2010) but also post-IPO M&A decisions. Fig. 5c and d show out-of-sample prediction of post-IPO bidders and targets, respectively. To this end, we first run our baseline estimations using data from 1985 up to 2000 to obtain parameter estimates from the Cox model. Using these parameter estimates and variables observed from the data between 2001 and 2008, we construct fitted likelihood of firms to become bidders or targets. The out-of-sample prediction rates of 65.9% for bidders and 63.1% for targets still provide a substantial improvement over randomness.

3.4. Financing post-IPO acquisitions

In this section, we present some evidence that deal structure variables determined at the time of IPO affect not only the likelihood of post-IPO mergers, but also financing methods of these mergers. In Table 6, we investigate to what extent these IPO deal structure variables have an impact on the total volume of acquisitions, and the amount of cash-financed and stock-financed acquisitions of IPO issuers in our dataset. Our results indicate that *Underpricing* is positively associated with the size of the acquisitions and the volume of stock-financed acquisitions. This finding is consistent with the hypothesis that underpricing helps

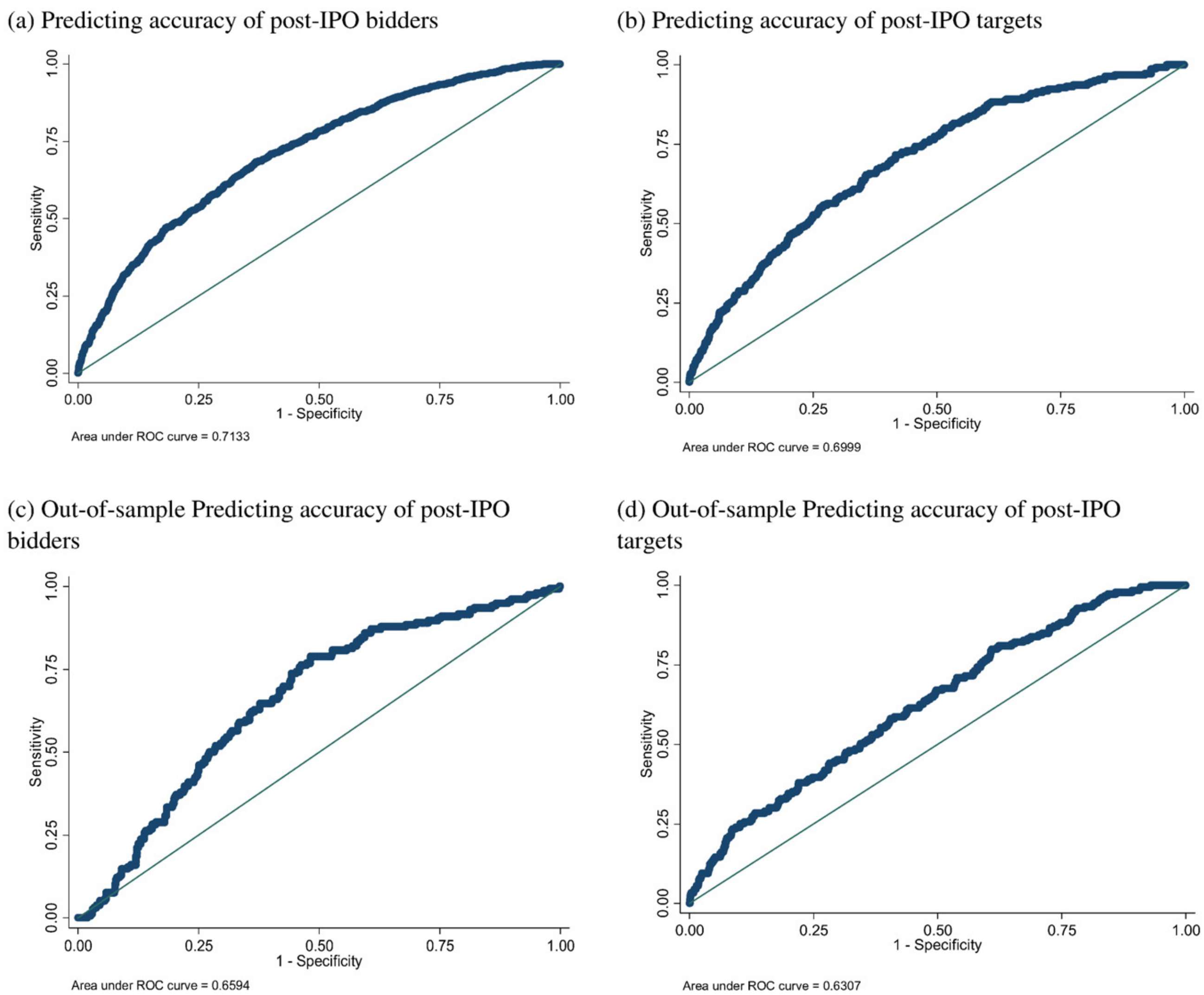

Fig. 5. Predicting accuracy of post-IPO bidders and targets. Panels a and b display the ROC curves to evaluate the accuracy of our model to predict post-IPO bidders and targets, respectively. Panels c and d display the ROC curves to evaluate the out-of-sample predicting accuracy of our models to predict post-IPO bidders and targets, respectively. Y-axis represents sensitivity, which is the proportion of acquisitions (Panels a and c) or sellouts (Panels b and d) accurately detected by our model, while X-axis represents inaccurate, false alarms generated by the model. Overall accuracy of our prediction models is shown in the area under the curves.

Table 6
Regressions of total, cash-financed and stock-financed acquisitions by post-IPO bidders.

	(1) Total volume of acquisitions	(2) Total volume of cash-financed acquisitions	(3) Total volume of stock-financed acquisitions
Pricing & proceeds			
Underpricing	0.2887*	−0.0129	0.2909*
	(0.0773)	(0.5898)	(0.0764)
Log(Primary Proceeds)	6.3818	1.0316	1.2894
	(0.3560)	(0.7820)	(0.8039)
Log(Secondary Proceeds)	−0.1702	−0.0996	−3.0030**
	(0.9369)	(0.8563)	(0.0441)
Log(Overallotment Proceeds)	−3.0668	2.8906	−4.8436
	(0.7136)	(0.1712)	(0.5225)
Price Revision	−0.2269	0.0045	−0.1257
	(0.1596)	(0.8835)	(0.3780)
Disclosure Dummy	4.9145	1.1895	0.5174
	(0.6619)	(0.5101)	(0.9563)
Post-IPO 20-day Return	16.1106	2.3262	11.1463
	(0.2367)	(0.4169)	(0.3436)
Ownership structure			
Overhang	−1.4120	−0.0884	−1.1202
	(0.4272)	(0.6139)	(0.5033)
IO Ratio	−114.7607*	9.3157	−119.6168*
	(0.0885)	(0.3526)	(0.0609)
IO Breadth	2.2304**	−0.0557	1.7783*
	(0.0377)	(0.5743)	(0.0880)
Institutional Ownership HHI	4.6788	−0.9007	2.8871
	(0.5779)	(0.3304)	(0.7155)
Dual Class Dummy	−14.7389	−1.3442	−17.0692*
	(0.2359)	(0.6764)	(0.0831)
Promotion & underwriting			
All-star Analyst Dummy	10.9371	−1.7168	17.8257
	(0.5408)	(0.4302)	(0.2949)
Bid-ask Spread Growth	0.0951	−0.0128	0.0962
	(0.4643)	(0.1721)	(0.4560)
Bookrunner Rank	−1.6315	−0.1076	−0.0146
	(0.1530)	(0.8061)	(0.9866)
Market condition & growth strategy			
Pre-IPO Sentiment	−0.8625	−4.7837	8.6945
	(0.9573)	(0.1871)	(0.5422)
Industry Acquisition Intensity	−1.9473	−6.1289	3.1658
	(0.9781)	(0.5642)	(0.9632)
Organic Growth/Assets	4.0074	−7.8488**	13.8418
	(0.9021)	(0.0111)	(0.6525)
Pre-IPO Acquisition Dummy	−0.4952	6.0215	−18.5271
	(0.9754)	(0.3519)	(0.1378)
Control			
VC Dummy	5.5486	−0.6091	6.5689
	(0.3255)	(0.5070)	(0.1968)
Age	0.0964	0.1539**	−0.1985
	(0.5515)	(0.0260)	(0.1435)
Market-to-Book	3.9440*	0.1640	3.5105*
	(0.0580)	(0.3001)	(0.0754)
ROA	5.5741	−0.3273	8.2675
	(0.6599)	(0.8503)	(0.4731)
Book Leverage	69.3384***	10.0290***	39.3148*
	(0.0058)	(0.0004)	(0.0981)
Constant	−72.9766**	−8.6785	−22.1844
	(0.0454)	(0.6475)	(0.3632)
Industry fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Observations	2993	2993	2993
R-squared	0.103	0.061	0.086

This table presents OLS regressions estimating total, cash-financed, and stock-financed acquisition volume within one year post IPO. Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008, and we track IPO firms' acquisition activity for up to three years post-IPO. Specifications (1) to (3) regress total volume of acquisitions, total volume of cash-financed acquisitions, and total volume of stock-finance acquisitions on IPO deal structures, respectively. Each estimation includes a set of control variables such as VC back dummy, firm age, market-to-book, ROA, book leverage, and firm fixed effects and industry fixed effects based on the revised FF 49 industry classification. P-values based on robust standard errors are reported in parentheses, and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively. Proceeds and assets are expressed in 2003 dollars. All IPO variables are defined in Appendix A.

generate acquisition currency and increase the liquidity of stocks in the secondary-market to issue and swap shares for financing prospective acquisitions. In contrast, organic growth via investment in R&D and capital expenditure and external growth via cash-financed acquisitions are negatively correlated, consistent with the argument that newly public firms tend to follow internal or external growth strategies. Finally, newly public firms that allocate more shares to institutional investors rely relatively less on stock financing for merger bids.

4. Institutional investment and post-IPO merger activity

Given the frequency and its important value implication, post-IPO acquisitions or sellouts can be critical for investors' investment decision in firms during the IPO and post IPO period. If investors have an ability to distinguish potential acquirers and targets from firms which do not get involved with the M&A market, then they might tilt their exposure toward or away from firms based on their likely future M&A activity. There is general consensus in the literature that some institutional investors are skilled at processing public information. In particular, Field and Lowry (2009) find that their superior investment performance in IPO firms appears attributed to their ability to read and interpret publicly available information at the time of IPO. We posit that if a future post-IPO merger event is indeed predictable on the basis of firm and IPO deal characteristics at the time of IPO, this prediction ought to manifest itself more readily in institutional investors' investment behaviors.

We investigate if institutional investors make use of information observable at the time of IPO to interpret the merger tendency of newly public firm. We collect the institutional holding data on the Spectrum reporting date at least 4 weeks after the IPO to measure voluntary post-IPO holdings by each institution. Institutional investors purchase a fairly constant percentage (about 70%) of the shares offered in IPOs, but they flip heavily the shares that they receive within the first month (Hanley and Wilhelm, 1995; Ljungqvist and Wilhelm, 2002; Aggarwal, 2003; Chemmanur and Hu, 2007). A period of at least four weeks ensures that institutional holding measure reflects institutions' willingness to hold the position in a stock rather than the initial allocation that institutions receive or an institutions' inability to move in or out of a position in a newly public firm. We are interested in whether institutional investors are making use of the publically available information, similar to what we include in our prediction model, or the private information in their identification. To measure the expected institutional ownership based on public information measures, we estimate a fractional logistics regression of total institutional ownership on the public information measures listed in Table 2 except ownership structure variables.[11] The difference between actual institutional ownership and expected institutional ownership is the measure of unexpected institutional investment, i.e., institutional ownership purged of some publicly available information. Hence, the unexpected institutional ownership is a combination of private information and public information omitted from our specifications. As shown in Table 7, we find that unexplained institutional ownership has little explanatory power in predicting post-IPO M&A activity. This finding suggests that at the time of IPO institutional investors seem to rely primarily on publicly available information to invest in potential M&A participants.[12]

Next, while sophisticated institutional investors, in general, may spot good firms at the time of IPOs which subsequently engage in M&A activities, we exploit the fact that different types of institutional investors fundamentally vary in their investment behaviors. Transient institutional investors are known to be short-term traders and rely heavily on hard, tangible, publicly available information, but dedicated institutional investors rely relatively more on soft, intangible information (Bushee, 2004). Fig. 6 displays transient and dedicated institutional ownership at likely bidders over twelve quarters post IPO based on our prediction models. We classify likely bidders as IPO firms with above median predicted probability to become a bidder based on the Cox hazard model in Table 4. The construction of dedicated or transient ownership follows Bushee's (2001) classification. Fig. 6 depicts that the gap in institutional ownership at likely bidders versus unlikely bidders widens during the post IPO period. However, within first quarters of IPOs, dedicated institutional ownership at likely bidders versus unlikely bidders is almost indistinguishable, while transient institutional ownership at likely bidders is around 2% larger than that at unlikely bidders.

To get better insight into the question whether institutional investors, particularly transient institutional investors, are able to discern acquisition activity *realized* in the near future, we implement a Propensity Score Matching approach. Using one-to-one matching approach, we match IPO firms which eventually became bidders with other IPO firms based on a wide array of firm characteristics *except for* IPO deal characteristics. Firm characteristics variables used for matching are market-to-book, R&D and capital expenditures, book leverage, cash ratio, industry classification, IPO year and year-time dummies. Given the large frequency of IPOs and post-IPO mergers in our dataset, our matching approach lets bidder and non-bidder IPO firms be similar to a large extent in key firm characteristics but differ from one another in terms of IPO deal characteristics. Having matched IPO bidder and non-bidder firms on a one-to-one basis, we conduct a difference-in-difference (DID) estimation to examine how the institutional investments in IPO bidders and matched IPO firms varied before and after the first merger announcements over the next 4 years following IPOs.[13] The premise is that if institutional investors are adept at reading publicly available information at the time of IPO, then even before acquisition announcements, they should be able discern newly public firms which eventually

[11] Consistent with Papke and Wooldridge (1996), we estimate a fractional response of institutional ownership ratio, assumed to range between 0 and 1 interval via a quasi-likelihood approach of a fractional logit model.

[12] These results also help address a related concern of information leakage. That is, investors may learn M&A deals from the information leakage right before the merger announcement (leading to stock price run-up), which contributes to the insignificant announcement returns for likely bidders. Though this possibility is hard to test directly and we acknowledge the possible information leakage right before the merger announcement, we find in Table 7 that institutional ownership explained by the information disclosed at the time of IPO predicts the post-IPO merger activity.

[13] We limit our DID estimations to bidder and non-bidder IPO firms because institutional ownership at target firms cannot be observed without noise after some time following takeovers as they are dissolved into acquirers.

Table 7
Expected Institutional Ownership.

Dependent variable	(1) Bidder	(2) Target	(3) Bidder	(4) Target
IO Ratio (30 days post IPO)	1.6191***	2.1800***		
	(0.0078)	(0.0002)		
Expected IO Ratio			1.4605**	1.8676***
			(0.0400)	(0.0036)
Unexpected IO Ratio			1.2718	1.4097
			(0.2213)	(0.1330)
Controls	Yes	Yes	Yes	Yes
Observations	2993	2993	2993	2993

This table presents the proportional Cox hazard regression estimating the relation between institutional ownership and the probability of post-IPO merger activity. Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008. Total IO is institutional holdings as a percent of the public float. Institutional holdings are measured at the first reporting date occurring 4 weeks post-IPO. To measure the Expected IO based on public information measures, we estimate a fractional logistic regressions of total institutional ownership on the public information measures listed in Table 4 or 5 except ownership structure variables. We name the fitted value from the fractional logistics regression as "Expected Institutional Ownership" (*Expected IO Ratio*). The difference between actual institutional ownership and expected institutional ownership is the measure of "Unexpected Institutional Ownership" (*Unexpected IO Ratio*) - that is, "institutional ownership" purged of some publicly available information. All other control variables in Table 4 or Table 5 are included in the estimation but not reported for brevity. P-values based on robust standard errors are reported in parentheses, and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively.

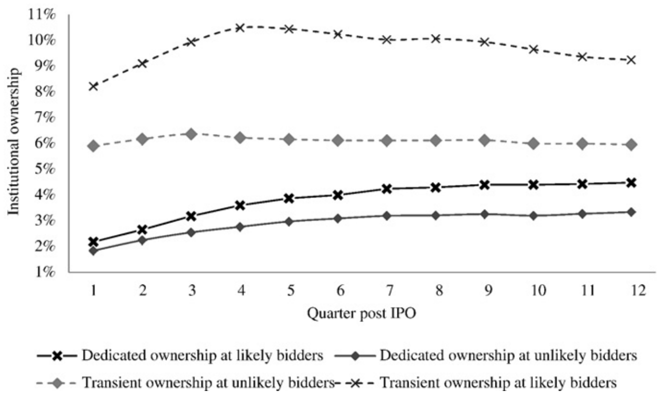

Fig. 6. Institutional ownership at likely/unlikely bidders. This figure displays institutional ownership of likely and unlikely bidders over 12 quarters post IPO. Dedicated or transient ownership follows Bushee (2001) classification. Likely bidders are IPO firms with above median predicted probability of the firm becoming a bidder or a target based on the Cox hazard model in Table 4.

become bidders. This prediction ability is expected to be particularly stronger for transient institutional investors. We focus on four years post-IPO period to allow for variations in institutional ownership at least 1 year after a post-IPO merger which may take place within 3 years of IPO. The results are displayed in Table 8.

Panel A presents the cross-sectional differences in institutional ownership at bidder firms and propensity-score-matched non-bidder IPO firms. The findings from Panel A suggest that institutional ownership at newly public bidder firms tend to be significantly larger than matched non-bidder firms cross-sectionally over the four years post IPO. Even though the results from Panel A point to a tendency of institutional investors, particularly transient investors, to invest more in IPO firms that eventually became bidders compared to non-bidders, the results do not necessarily suggest that institutional investors were significantly more present at IPO bidder firms before merger announcements were made.

Based on the matched sample, Panels B and C present DID estimations of ownership of transient and dedicated institutional investors, respectively, at matched IPO bidder and non-bidder firms before and after the first post-IPO merger announcements. *Bidder* is a post-IPO bidder dummy, which equals 1 if a newly public firm becomes a bidder within 4 years of its IPO, 0 otherwise. *Post M&A* is a post-merger period dummy, which equals 1 if the time of observation is after the first acquisition event. The interaction term, *Bidder* × *Post-M&A*, is the DID term. Since whether or not an IPO firm becomes an eventual bidder is constant over the estimation period, we employ random effect panel estimations in lieu of fixed effect estimations and cluster standard errors at the firm level.[14] The findings from Panel B show that compared to matched non-bidder IPO peers, IPO bidder firms attract significantly more transient institutional investors even before the merger announcements are made. While transient institutional ownership is, on average, around 6.3% at matched non-bidder firms, this figure is close to 7.8% at bidder firms before mergers. The

[14] Available upon request, pooled OLS estimations yield similar results.

Table 8
Institutional ownership at bidder- and non-bidder newly public firms during post-IPO period.

Panel A: one-to-one matching (with replacement)	(1) Control group (non-bidder)	(2) Treatment group (Bidder)	(3) Mean difference (2)–(1)	(4) *t*-Stat
Transient IO Ratio	0.0603	0.0809	0.0206***	**8.97**
Dedicated IO Ratio	0.0174	0.0330	0.0156***	**15.46**

Panel B: transient institutional ownership	(1)	(2) Estimated partial derivatives evaluated at different values of *Bidder* and *Post-M&A* dummies.	(3)	(4)	(5)
		Post-M&A = 0, *Bidder* = 1	*Post-M&A* = 0, *Bidder* = 0	*Post-M&A* = 1, *Bidder* = 1	*Post-M&A* = 1, *Bidder* = 0
Bidder	0.0153***	0.0777***	0.0624***	0.0863***	0.0591***
	(0.0000)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Post-M&A	−0.0033				
	(0.2494)				
Bidder × *Post-M&A*	0.0119***				
	(0.0002)				
Constant	0.0624***				
	(0.0000)				
R-squared (within)	0.0174				
Observations	65,137				
Number of firms	4235				

Panel C: dedicated institutional ownership	(1)	(2) Estimated partial derivatives evaluated at different values of *Bidder* and *Post-M&A* dummies.	(3)	(4)	(5)
		Post-M&A = 0, *Bidder* = 1	*Post-M&A* = 0, *Bidder* = 0	*Post-M&A* = 1, *Bidder* = 1	*Post-M&A* = 1, *Bidder* = 0
Bidder	0.0024*	0.0285***	0.0261***	0.0386***	0.0291***
	(0.0590)	(0.0000)	(0.0000)	(0.0000)	(0.0000)
Post-M&A	0.0031*				
	(0.0665)				
Bidder × *Post-M&A*	0.0070***				
	(0.0002)				
Constant	0.0261***				
	(0.0000)				
R-squared (within)	0.0114				
Observations	65,137				
Number of firms	4235				

Using one-to-one propensity score matching, IPO bidder firms are matched with non-bidder IPO firms based on a wide array of firm characteristics except for IPO deal characteristics. Firm characteristics variables used for matching are market-to-book, R&D and capital expenditures, book leverage, cash ratio, industry classification, IPO year and year-time dummies. Panel A shows the cross-sectional difference in institutional ownership at matched bidder and non-bidder IPO firms during the post-IPO period. Panels B and C present difference-in-differences estimations of ownership of transient and dedicated institutional investors in matched IPO bidder and non-bidder firms before and after the time of first post-IPO merger announcements in a panel data. *Bidder* is a post-IPO bidder dummy, which equals 1 if a newly public firm becomes a bidder within 4 years of its IPO, 0 otherwise. *Post M&A* is a post-merger period dummy, which equals 1 if the time of observation is after the first M&A event. The interaction term, *Bidder* × *Post-M&A*, is the DID term. The estimations focus on the first M&A event within 4 years of IPOs. Since whether or not an IPO firm will become an eventual bidder is fixed over the estimation period, random effect within-estimations are employed instead of fixed effect estimations. Standard errors are clustered at the firm level. P-values are shown in parentheses, and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively.

difference in transient ownership is about 1.5%. The gap in transient institutional ownership between matched non-bidder and bidder IPO firms gets larger after merger announcements, as the positive and statistically significant DID coefficient suggests. Panel C presents the DID estimation of dedicated institutional ownership at newly public bidders and matched non-bidders. First, dedicated ownership is much smaller at both IPO non-bidders and bidders at around 2% to 3%. Second, although compared to matched non-bidder IPO firms, IPO bidder firms attract more dedicated institutional investors before mergers, the difference is statistically and economically weak. Once again, the positive and statistically significant DID estimator suggests that the gap in dedicated institutional ownership grows post mergers. Overall, the DID analyses on realized post-IPO acquisitions in Table 8 support the view that institutional investors, particularly transient investors, are able to distinguish newly public future bidders from their closely matched IPO peers even prior to mergers.

Results in Table 8 for *realized* bidders and Fig. 6 for *likely*/*unlikely* bidders derived from the prediction models provide evidence that particularly transient investors which are known to be superior in processing and interpreting hard, publicly available information and inherent IPO and firm characteristics anticipate future merger deals and incorporate their valuation effects before they are realized by hoarding significantly more shares of likely bidders to possibly exploit short-term trading opportunities.

5. Market reaction to merger announcements and merger characteristics

The results shown in the prior sections suggest that information observable to investors at the time of a firm's IPO can distinguish likely future M&A bidders and targets from the rest of the population of newly public firms and that institutional investors appear to act on such information. Given that post-IPO bidder and target activity is somewhat predictable, especially in the case of the more frequent bidding activity observed among newly public firms, a natural question is whether stock-price reaction to announcements of M&A activity is conditioned on anticipation of such activity.

To address this question we calculate the five-day cumulative abnormal returns (CARs) over the prior 2 days through the subsequent 2 days following the announcement of the first M&A activity involving a newly public firm as either a bidder or target. Since targets are usually acquired at a premium to the pre-acquisition stock price, target announcement returns are positive and large. The unconditional mean (median) abnormal return for our sample of newly public bidders is also positive. This appears to run counter to the commonly accepted wisdom (e.g., Bruner, 2004) that on average targets win and bidders don't gain and may lose. However, as noted by Netter et al. (2011), that common wisdom is confined mostly to samples of large firm acquisitions of public targets that have been subject to a variety of data screens. They find that in contrast to the commonly accepted wisdom, the unconditional average return to acquisition bidders, especially small bidders such as those in our sample of newly public firms, is positive.

We propose two competing hypotheses, the anticipation hypothesis versus the preparation hypothesis, for the effect of M&A likelihood on observed announcement returns for newly public firms that engage as bidders or that are targeted for acquisition. The anticipation hypothesis would suggest that since M&A activity is expected, more likely bidders and more likely targets, in particular, would have the positive gain from M&A activity already reflected in their stock prices. Consistent with this hypothesis, we expect that M&A announcement returns would be less positive for more likely bidders and more likely targets. This is broadly consistent with Cai et al.'s (2011) findings that less anticipated merger bids earn significantly higher announcement returns, as well as Schipper and Thompson (1983) and Malatesta and Thompson (1985) who show that bidder returns are attenuated for program acquirers. An alternative hypothesis, the preparation hypothesis, would suggest an opposite finding of more positive returns for more likely bidders and targets. An unlikely bidder may be a firm that has not properly structured or well-timed its IPO to make efficient and well-priced acquisitions subsequently. Consequently, the market may not react sanguinely to a bid announcement by such a firm, and thus the announcement return by likely bidders may exceed those of unlikely bidders. Similarly, a newly public firm that appears an unlikely target may have ended up with dissatisfied shareholder base, unfavorable underwriter support, or management frustrated by the inability to pursue a path to sell the firm via a takeover. Unlikely targets may therefore command lower takeover premiums and experience lower returns upon being announced as a target of an acquisition. This

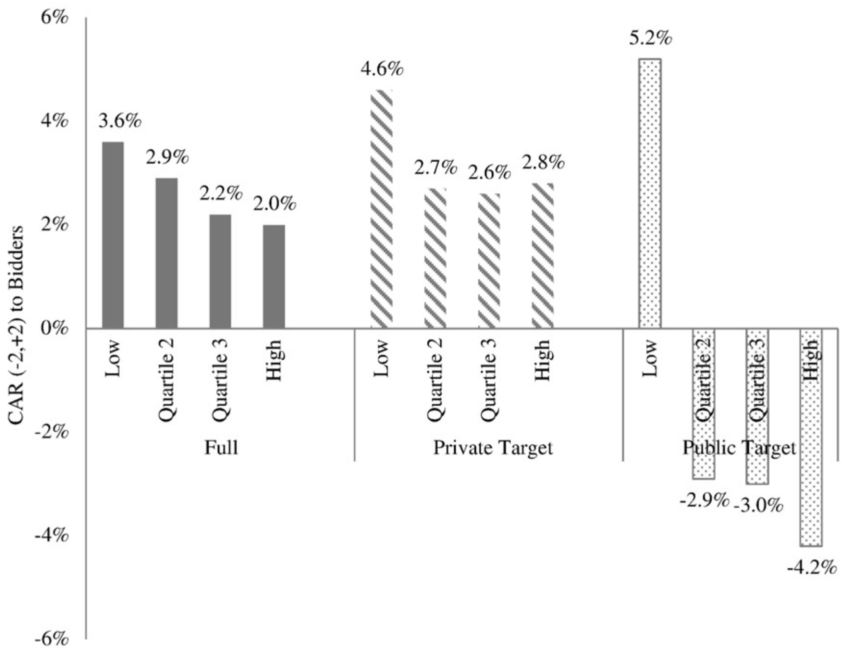

Fig. 7. Bidder announcement CARs (−2,+2) based on the probability of becoming a bidder implied by the Cox Hazard model in Table 4. This figure shows the cumulative abnormal returns on the common stock of newly public bidders over the 5 days surrounding each firm's first announcement of a post-IPO takeover bid. The quartile assignments are made unconditionally for the entire sample of newly public firms based on estimated bidder likelihood, not merely for those firms that eventually become bidders.

reflects Banerjee and Owers' (1992) argument that bidders prepared for acquisitions in terms of their growth and value creation strategies benefit from the preparation.

5.1. Short-term returns to newly public bidders and targets

Fig. 7 shows the cumulative abnormal returns on the common stock of newly public bidders over the 5 days surrounding each firm's first announcement of a post-IPO takeover bid across four quartiles of predicted probability of the firm becoming a bidder based on the Cox model in Table 4.[15] The mean return in the lowest quartile in terms of bidder probability is positive 3.6% (P-value < 1%). In contrast, announcement returns across the other three quartiles of increasing bidder likelihood are substantially decreasing with 2.0% for the highest quartile. This evidence is consistent with the anticipation hypothesis, suggesting that investors react less positively to bids by likely bidders because growth through acquisition is expected to some extent and hence already priced into the firm's common shares. M&A announcement returns are thus more favorable for relatively less likely bidders. We further split the full sample into the private target group and the public target group. We observe that although announcement CARs are also decreasing for the private target group, the decreasing pattern is much more striking for the public target group. For acquisitions of public targets, the mean return in the lowest quartile in terms of bidder probability is positive 5.2%, while announcement returns across the other three quartiles are all negative and decreasing in the probability of becoming a bidder.

Fig. 8 shows short-term market reactions in target share price when sample firms become targets of acquisitions across quartiles based on target probability. Target CARs are positive and materially different from zero across each quartile, and become the lowest for the target firms in the highest quartile of target probability. On average, the more anticipated the target firm, the lower the cumulative abnormal return.

Table 9 shows an ordinary least squares regression in which the dependent variable is the CARs upon the announcement of the first merger bid by a newly public firm within our sample. We regress the announcement CAR to the newly public bidder on the predicted probability of the firm being a bidder and several other variables observed at the time of the merger announcement. The other variables include the percent of the deal financed by cash, public target dummy, diversification dummy, tender offer dummy, acquirer market value of equity 28 days before the merger, deal value relative to bidder size, the percent of the target firm acquired, time-to-M&A, whether IPO underwriters are hired as merger advisors, year fixed effects, and industry fixed effects. As shown in column (1), the bidder CAR is decreasing in the probability of the firm being a bidder based on variables observable at the time of its IPO, consistent with the pattern that emerged in Fig. 7. We also interact probability of being an acquirer with *Public Target dummy*, and *Underwriter/AD Dual Dummy*. The relation between bidder CARs and *Pr(Bidder)* is primarily driven by public targets group as the interaction term is negative and significant at the 1% level. We argue that acquisitions of public targets are more transparent to investors in terms of merger expectation and valuation. Given the same level of probability of becoming a bidder based on deal structure variables observed at IPO, investors are able to gather more information and more accurately predict the probability and valuation of acquisitions. As a consequence, acquisition announcement returns are substantially lower with more valuation realized at IPO or prior to the merger. However, these univariate patterns do not hold for target firms after controlling for other merger and firm characteristics in the specifications (3) and (4) of Table 9.

Taken together, evidence from Figs. 7 and 8 and Table 9 shows that investors, on average, favorably respond to relatively unanticipated, surprise merger announcements for both acquirers and targets. Our findings confirm the literature (e.g., Netter et al., 2011) that while announcement returns to bidders of public large targets are consistently negative, the unconditional average returns to bidders, especially small bidders of private targets are positive. Merger announcement CARs to newly public firms are decreasing in the probability of becoming a bidder perhaps because growth prospects via acquisitions have already been priced and incorporated into equity markets, especially when the target is a public firm, making it easier for the market to analyze and forecast. We find that unanticipated acquisition announcements of private targets generate more positive ARs, as depicted in Fig. 7. Similarly, unanticipated, relatively surprising acquisition announcements of public targets generate less negative, or even positive ARs. We believe that these findings consistent with our anticipation hypothesis contribute to the understanding of the broad controversy in the literature concerning investor reactions to bidders' acquisition announcements.[16]

[15] The quartile assignments for Fig. 7 are made unconditionally for the entire sample of newly public firms based on estimated bidder likelihood, not merely for those firms that eventually become bidders. Consequently, the population from each quartile of firms that do indeed become bidders is increasing from the low to high quartiles. The quartile assignments for Fig. 8 are similarly made for all firms based on target likelihood, and hence the number of firms that indeed become targets is increasing from the low to high quartile.

[16] Houston et al. (2001), for instance, attempt to understand why stock return evidence fails to find positive gains from acquisitions of bidders even if they exist. Houston et al. (2001) list the following reasons: (1) merger announcements mix information concerning the proposed acquisition with information about the financing of the acquisition. Stock offerings are generally interpreted as signals of the bidder's "overvaluation"; (2) disappointment that the bidding firm is less likely to be acquired in the future; (3) acquisitions are largely anticipated and already priced. Hence, CARs around merger announcements may represent other mixed signals. Clearly, our study is directly related to (3). The results show that unanticipated acquisitions generate relatively larger, even positive CARs for bidders, consistent with Houston et al. (2001).

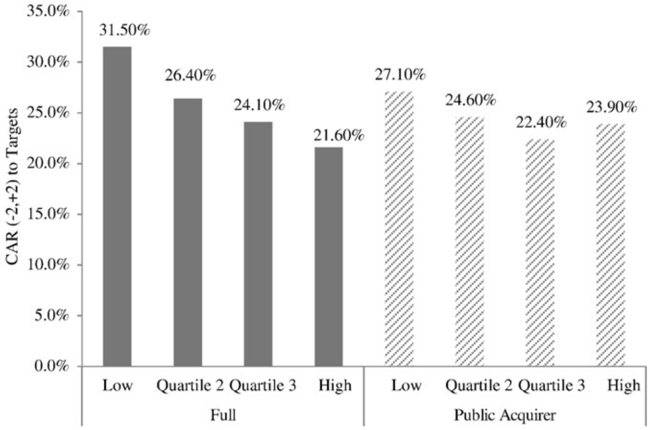

Fig. 8. Target announcement CARs (−2,+2) based on the probability of becoming a target implied by the Cox Hazard model in Table 5. This figure shows the cumulative abnormal returns on the common stock of newly public targets over the 5 days surrounding each firm's first announcement of a post-IPO takeover bid. The quartile assignments are made unconditionally for the entire sample of newly public firms based on estimated target likelihood, not merely for those firms that eventually become targets.

5.2. Long-term returns to newly public bidders

If a newly public firm which is not anticipated to be a bidder in the short-run releases an acquisition announcement, unexpected ("surprise") information may result in investor overreaction priced at the time of announcement. Unlikely bidder-IPO firms may then experience a long-term return reversal following their unanticipated acquisition announcements. On the other hand, for IPO firms which are largely anticipated to be bidders, there should not be significant surprise element triggering investor overreaction upon acquisition announcements, thereby having no basis for long-term reversal.

We test these conjectures and show results in Table 10. The results in Panel A show that for one-year post-announcement returns Jensen's alpha is negative for unlikely bidders, and not different from zero for likely bidders. Panels B and C focus on two-year and three-year long-term returns and show similar results, suggesting that unlikely bidders, although they experience positive short-term announcement returns, experience long run underperformance. These findings are consistent with the anticipation hypothesis and provide new insights to the extant literature. It is a very well-known "puzzle" that newly public IPO firms, on average, underperform in the long-run (Ritter, 1991; Loughran and Ritter, 1995; Ritter and Welch, 2002). Brau et al. (2012) show that post-IPO acquisitions sink IPO performance. We contribute to this strand of literature by showing the post-IPO long-run underperformance is driven in part by relatively unanticipated bidders.

6. Conclusion

M&A activity among newly public firms is ubiquitous, and links between the going public process and subsequent M&A activity have been acknowledged by prior research.[17] We investigate how IPO-specific variables observed at the time of a firm's IPO affect the likelihood and timing of M&A, institutional ownership in newly public firms based on likely M&A activity, and stock-pricing consequences of post-IPO M&A activity.

Our various prediction models which focus on 5927 U.S. IPOs from 1980 to 2008 generate interesting results. First, public information underlying various aspect of IPO deal structure signaled at the very beginning of firms' public life is predictive of future M&A activity. In particular, we find that the likelihood that a newly public firm becomes an acquisition bidder versus a target within a few years of its IPO is influenced by aspects of IPO pricing, structure of proceeds, resulting post-IPO ownership structure, and the underwriting and promotional process. We show that these inherent IPO characteristics and information signaled at the time of IPOs help predict newly public firms' future M&A activities approximately 70% of the time. These results confirm that firms do indeed go public to acquire or to be acquired in a manner that links these ultimate strategic objectives to information observable at the time of the IPO. Second, our findings on institutional ownership suggest that sophisticated investors seem to rely on information observable at the time of the IPO to invest in potential M&A participants. Stock holdings of transient institutional investors, in particular, suggest that such investors distinguish IPO bidder firms from non-bidder IPO firms prior to merger announcements. Third, we find that firm characteristics observed at the time of the IPO condition stock-price changes upon announcement of M&A activity subsequent to the IPO. Specifically, announcement returns for bids by newly public firms are decreasing in anticipated bidder likelihood based on variables observed ex ante at the time of the IPO. The positive short-term abnormal returns

[17] Hovakimian and Hutton (2010) provide a parsimonious review of prior literature that links IPO and M&A activity, including Zingales (1995), Mello and Parsons (1998), Schultz and Zaman (2001), Rosen et al. (2005), and Rau and Stouraitis (2011).

Table 9
OLS regression of merger announcement CARs.

	(1) Bidder CAR(−2 + 2)	(2) Bidder CAR(−2 + 2)	(3) Target CAR(−2 + 2)	(4) Target CAR(−2 + 2)
Pr(Bidder)	−0.0047*	−0.0035		
	(0.0825)	(0.4886)		
Public Target Dummy	−0.0519***	0.0193		
	(0.0001)	(0.6491)		
Pr(Bidder) ∗ Public Target Dummy		−0.0376**		
		(0.0295)		
Pr(Bidder) ∗ UW/AD Dual Dummy		0.0305*		
		(0.0991)		
Pr(Target)			0.0003	0.0367
			(0.6934)	(0.0264)
Public Acquirer Dummy			−0.0889**	0.0383
			(0.0303)	(0.6714)
Pr(Target) ∗ Public Acquirer Dummy				−0.0561
				(0.1313)
Pr(Target) ∗ UW/AD Dual Dummy				0.0348
				(0.1224)
UW/AD Dual Dummy	0.0176	−0.0169	−0.0212	−0.0625
	(0.3251)	(0.1444)	(0.435)	(0.2824)
Acquirer MVE	−0.0001	−0.0001		
	(0.2414)	(0.2341)		
Target MVE			-0.0001***	−0.0001***
			(0.0000)	(0.0000)
Time-to-Merger	−0.0001	−0.0001	0.0001	0.0001
	(0.8554)	(0.8484)	(0.4921)	(0.5234)
% of Cash Consideration	0.0001	0.0001	0.0013***	0.0014***
	(0.3280)	(0.3312)	(0.0003)	(0.0003)
Diversification Dummy	0.0012	0.0005	−0.0147	−0.0167
	(0.3041)	(0.3234)	(0.3651)	(0.3398)
Tender Offer	−0.0224	−0.0110	0.0587*	0.0267**
	(0.6293)	(0.6265)	(0.0592)	(0.0490)
% Acquired	0.0001	0.0008	0.0017	0.0018
	(0.2547)	(0.4914)	(0.2163)	(0.2089)
Relative Size	0.0047	0.0045		
	(0.3110)	(0.3168)		
Constant	0.0094	−0.0785	0.1739	−0.3932
	(0.1236)	(0.1973)	(0.2811)	(0.7243)
Industry fixed effects	Yes	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes	Yes
Observations	1668	1668	1124	1124
R-squared	0.042	0.046	0.134	0.136

This table presents the OLS regression estimating merger announcement CARs. Our IPO sample consists of 5927 U.S. IPOs from 1980 to 2008. Columns (1) to (4) present the OLS estimations where cumulative abnormal returns around the 5 days of merger announcements are regressed on predicted probability of becoming an acquirer or a target. The predicted probabilities of becoming a post-IPO bidder, *Pr(Bidder)*, and the predicted probability of becoming a post-IPO target, *Pr(Target)*, are calculated from the Cox hazard models in Tables 4 and 5. Public target dummy equals 1 if the newly IPO issuer acquires a publicly held target firm, and 0 otherwise. Public acquirer dummy equals 1 if the newly IPO issuer is acquired by a publicly held bidder, and 0 otherwise. *UW/AD Dual Dummy* equals 1 if the newly IPO firm continues to use its IPO underwriter as its merger advisor, and 0 otherwise. *Acquirer MVE* is the bidder's market value of equity measured 28 days before the merger announcement. *Target MVE* is the target's market value of equity measured 28 days before the merger announcement. *Time-to-M&A* is the total number of days elapsed from the date of IPO to the announcement of an M&A deal. *% of cash considerations* is the percentage of cash used to finance the total deal volume. *Diversification dummy* equals 1 if the bidder and the target share the same 2-digit SIC code. *Tender offer* equals 1 if the deal is a tender offer, and 0 otherwise. *% Acquired* represents the percentage of the ownership that is transferred to the bidder after the merger. *Relative size* is the deal value of the target firm as a proportion of market value of the bidder. Since all of the bidders which acquire newly public firms are not public, we omit relative size in Columns (3) and (4). P-values based on robust standard errors are reported in parentheses, and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively. Proceeds and assets are expressed in 2003 dollars. All IPO variables are defined in Appendix A.

for unlikely bidders, however, are followed by a reversal in long-term returns. An overreaction pattern characterized by a reversal in long-term returns observed for unlikely bidders but not likely bidders provides additional evidence for the investor anticipation hypothesis.

These results have implications for IPO investors and academic studies on samples of newly public firms, especially firms likely to engage in growth by acquisition. Foremost, a wide array of IPO deal structures reflecting firms' motives for future corporate control may help investors make effective investment strategies taking into consideration not only the likelihood and but also timing of merger events. Next, returns around merger announcements by newly public firms should not be interpreted as unbiased estimates of the shareholder wealth effects of mergers, since for many firms M&A activity is predictable. Brau et al. (2012) provide a partial explanation to the puzzle of negative long-run post-IPO stock performance, showing that post-IPO acquisitions adversely affect future stock returns. Our findings show that the long-run post-IPO underperformance following acquisitions is driven by unanticipated acquisitions.

Table 10
Long-run post-acquisition returns of newly public firms.

	Intercept (%)	RMRF	SMB	HML	Adj. R-squared	No. of firms	Implied AR (%)
Panel A. One year							
Likely bidders	0.0100	1.1693***	0.9021***	−0.8078***	0.8543	1114	0.12
	(0.05)	(18.55)	(13.88)	(−9.37)			
Unlikely bidders	−0.5800**	1.2180***	1.0296***	−0.2765***	0.7273	758	−6.74
	(−2.00)	(16.18)	(12.82)	(−2.65)			
Panel B. Two years							
Likely bidders	−0.2700	1.2381***	0.8974***	−0.5998***	0.8216	1114	−3.19
	(−1.07)	(19.15)	(13.50)	(−6.88)			
Unlikely bidders	−0.7000***	1.2235***	1.0011***	−0.1492	0.7296	758	−8.08
	(−2.53)	(17.52)	(13.41)	(−1.56)			
Panel C. Three years							
Likely bidders	−0.2100	1.2658***	0.9288***	−0.4619***	0.8123	1114	−2.49
	(−0.86)	(20.01)	(14.25)	(5.45)			
Unlikely bidders	−0.5600**	1.1872***	0.9768***	−0.1254	0.7536	758	−6.52
	(−2.26)	(19.02)	(14.68)	(−1.49)			

This table shows the long-run post-IPO acquisition returns for likely bidder and unlikely bidder IPO firms. Long-run returns are estimated using Fama-French 3-factor model. The intercept is the Jensen's alpha. *RMRF* is the market premium. *SMB* is the small minus big firm premium. *HML* is the high book-to-market minus low book-to-market risk premium. Panel A focuses on long-run returns over 1 year following acquisitions. Panel B focuses on long-run returns over 2 years following acquisitions. Panel C focuses on long-run returns over 3 years following acquisitions. P-values are reported in parentheses, and asterisks next to coefficient estimates (*, **, and ***) correspond to 10%, 5%, and 1% significance, respectively.

Appendix A. Definition and data sources

Variable	Definitions and data sources
Underpricing	Percentage change from the offer price to the first day closing price *Thomson Financial's SDC New Issues, with fill-ins of missing data from CRSP*
Primary Proceeds ($million)	The capital (in millions) raised at the IPO from the sale of primary shares *Thomson Financial's SDC New Issues*
Secondary Proceeds ($million)	The capital (in millions) raised at the IPO from the sale of secondary shares *Thomson Financial's SDC New Issues*
Overallotment Proceeds ($million)	The capital (in millions) raised at the IPO from the sale of overallotment shares *Thomson Financial's SDC New Issues*
Price Revision	Percentage change from the middle of the original file price range to the offer price *Thomson Financial's SDC New Issues*
Disclosure Dummy	Equals one (zero otherwise) if the issuer discloses future acquisitions in the section of use of proceeds in the prospectus *Thomson Financial's SDC New Issues*
Post-IPO 20-day Return	Cumulative excess returns of the issuer in the (+1,+20) period after IPO *CRSP*
Overhang	Shares retained by pre-IPO shareholders as a percentage of shares offered *Thomson Financial's SDC New Issues*
Dedicated Institutional Ownership	Institutional holdings of dedicated institutional investors, per Bushee (2001), at the end of the quarter that occurs within 3 months of the IPO *Thomson-Reuters 13F, with institutional investor classification from Brian Bushee's web-site*
Transient Institutional Ownership	Institutional holdings (percent) held by transient and quasi-indexer institutional investors, per Bushee (2001) *Thomson-Reuters 13F, with institutional investor classification from Brian Bushee's web-site*
IO Breadth	The number of 13-F institutional owners filing at the end of the quarter that occurs within 3 months after the IPO *Thomson-Reuters 13F*
Institutional Ownership HHI	The sum of the squares of the institutional holdings at the end of the quarter that occurs within 3 months after the IPO *Thomson-Reuters 13F*
Dual Class Dummy	Equals one if the IPO firm has dual-class shares *Jay Ritter's web-site*
All-star Analyst Dummy	Equal one if the IPO is covered by an institutional investor all-star analyst (top3) from the bookrunners 1 year after IPO *Jay Ritter's web-site*
Bid-ask Spread Growth	Percentage increase in average quoted bid-ask spread of the first 20 trading days relative to the second 20 trading days *CRSP*
Bookrunner Rank	The Carter and Manaster (1990) ranking of the first-listed bookrunner *Jay Ritter's web-site*
Pre-IPO Sentiment	Baker-Wurgler Sentiment Index in the quarter prior to the IPO *Jeffrey Wurgler's web-site*
Industry Acquisition Intensity	Number of acquisitions in the quarter prior to the IPO divided by total firms in the Fama & French (1997) 49 industry *Thomson Financial's SDC Mergers&Acquisitions*
Organic Growth/Assets	Research and Development Expenditure and Capital Expenditure scaled by total assets in the IPO year *Compustat*
Pre-IPO Acquisition Dummy	Equals one (zero otherwise) if the issuer conducts any acquisition within 3 years prior to the IPO

(continued on next page)

(continued)

Variable	Definitions and data sources
	Thomson Financial's SDC Mergers&Acquisitions
VC Dummy	Equals one (zero otherwise) if the IPO was backed by venture capital
	Thomson Financial's SDC New Issues
Age	Calendar year of offering minus the calendar year of founding
	Jay Ritter's web-site
Book Leverage	Interest bearing debt scaled by assets ((Compustat item DLTT + item DLC) / item AT) in the IPO year
	Compustat
ROA	Operating income before depreciation to assets (Compustat item OIBDP / item AT) in the IPO year
	Compustat
Market-to-Book	Market value of assets to assets ((Compustat item AT − item CEQ + item CSHO ∗ item PRCC_F) / item AT) in the IPO year
	Compustat
Assets ($million)	Book value of total assets in the IPO year
	Compustat
Bubble Dummy (1999–2000)	Equals one (zero otherwise) if the IPO occurred during 1999–2000

Our sample consists of 5927 U.S. IPOs from 1980 to 2008. We exclude (a) the observations not identified on CRSP, (b) the observations in which the offering is not underwritten or is not available, (c) limited partnerships, (d) REITs, (e) closed-end funds, (f) spinoffs, (g) previous leverage buyouts, (h) units, (i) ADRs, (j) shares of beneficial interest, (k) foreign corporations, (l) banks and Savings and Loan (S&Ls), (m) best efforts offers, (n) SPACs, (o) IPOs with an offer price of less than $5.00 per share, (p) utility firms, and (q) IPOs which are not original. Proceeds and assets are expressed in 2003 dollars.

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